

Math Ma Workbook

Fall 2025

This workbook belongs to _____

If found, please contact (cell/email): _____

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UNIT 1

WORKING WITH FUNCTIONS, MEASURING THE AVERAGE RATE OF CHANGE

1.1 INTRODUCTION TO FUNCTIONS

1: Introduction to Functions

★ *Reminders:*

- If you haven't take the Ma/1a skill check, you will want to do that right away. Make sure to email our course head, Kate Penner at penner@math.harvard.edu if you haven't yet.
- If you haven't already, please sign up for Edfinity Math Ma 2025 ASAP after class to make sure it works on your device. It will host part of your homework assignment due at our next class!
- Purchase your workbook at Flashprint as soon as possible if you haven't already!
- Workshop time preferences are due today. They will run Tuesdays, beginning next week, Tuesday, September 16.

This worksheet is your personal guide to the material we cover in class. As you work through the problems, take some time to write down the key points you want to take away from class.

We intentionally make these worksheets too long to finish in class so that you'll always have some extra practice problems. Each section may cover different problems to cover learning objectives; we'll post solutions on the course website after class.

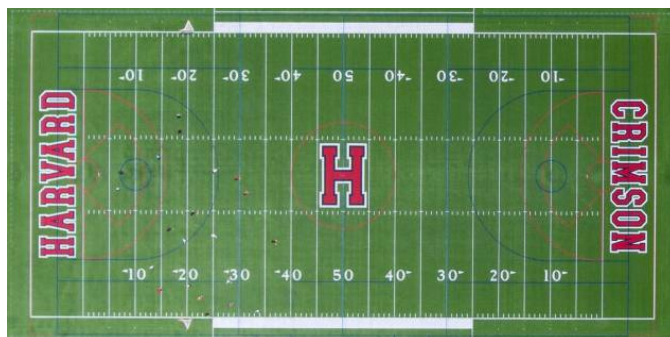
Each day, we'll list some learning objectives. Here are today's:

You should be able to:

- Explain what a *function* is, and distinguish between functions and non-functions.
- Identify the *independent variable*, *dependent variable*, *domain*, and *range* of a function.
- Understand the basics of function notation (naming and defining a function, identifying the input variable).
- Explain how slope graphically represents a rate of change.
- Reason about whether a function is *increasing* or *decreasing*, as well as whether it's *concave up*, *concave down*, or *linear*.

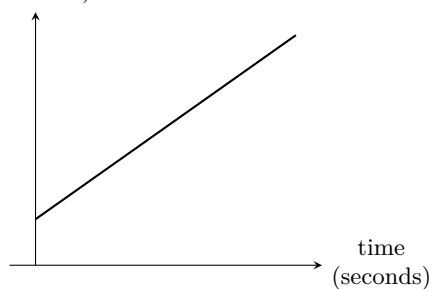
The learning objectives should give you an idea of the basic skills and concepts from each day that we expect you to be comfortable with. Class will introduce you to these ideas, but the pset is where you'll really get comfortable with them! After you finish the pset, take a look back at the learning objectives, and check how you're feeling about them. If you don't feel comfortable with an objective, we encourage you to stop by MQC or office hours to talk about it more.

1. a) During a football game at Harvard Stadium, the quarterback stands at the 50-yard line (right in the middle of the field) and throws the ball to a receiver near the end of the field marked CRIMSON.

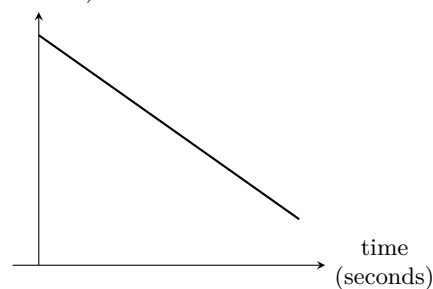


3 seconds after the quarterback releases the ball, the receiver catches the ball at the 10-yard line (which is labeled 10 in the picture). One of the following graphs shows the ball's horizontal position (which yard line it's at) while it's in the air. Which one?

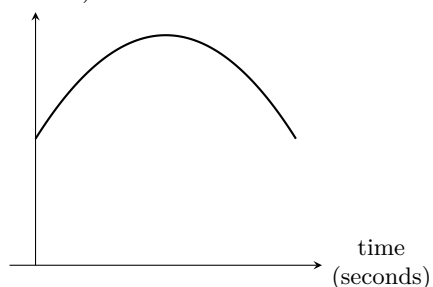
(A) position
(yard line)



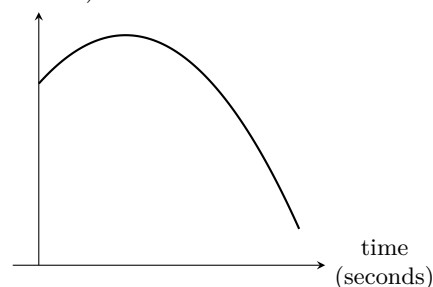
(B) position
(yard line)



(C) position
(yard line)



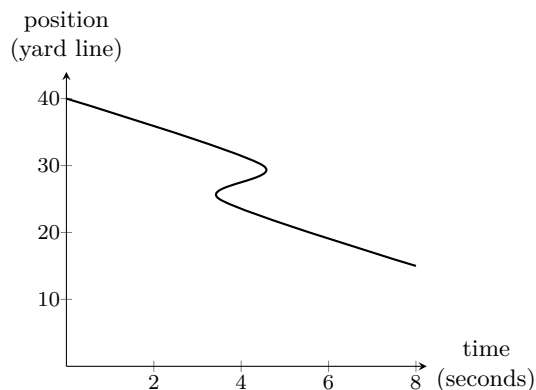
(D) position
(yard line)



Explain why you chose the graph you did.

- b) Sketch the graph you picked on the board. Label any values you can on the axes.

- c) Your friend Cris says, “I was watching another football game on TV; one of the players had the ball and ran with it for a while before he got tackled!” Cris draws the following graph to show you the ball’s position during this play. Another friend, Lisa, says, “I don’t think that graph could possibly describe where the ball was!” and asks you for your opinion. What do you think? How can you convince Cris and Lisa you’re right?



Some important terminology!

- A function is an input-output relationship with the key property that:
For each possible input, _____.
Graphically, this means _____.
- The input variable is called the _____ variable, and the output variable is called the _____ variable.
- When we graph a function, we put the _____ variable on the horizontal axis and the _____ variable on the vertical axis.
- The domain of a function is the set of all _____, and the range is the set of all _____.

Notice that both the output and input variables represent quantities whose values may vary.

- d) Look back at the function in (a), and identify the:

- independent variable
- dependent variable
- domain
- range

2. Harvard is ordering more colorful chairs for Harvard Yard. According to Flyby, the chairs cost \$381 each, with a minimum order of 2 chairs. Let $C(n)$ be the cost to buy n chairs.

a) Is C a function of n ? Why or why not?



b) What can you say about the domain of C ?

c) Sketch the graph of C , and explain how you came up with your graph.

A Note on Defining Functions and Function Notation

The same way we use variables to refer to varying quantities, we use function notation to more efficiently refer to a function and its related quantities.

Typically, we assign the function a name using a symbol, often a letter. In this course, the function name will often be the same as the output quantity's variable, such as C representing the cost of the chairs and the name of the function. In your own work or in another course, you may choose to name the output variable and the function separately.

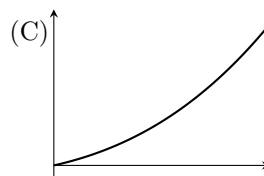
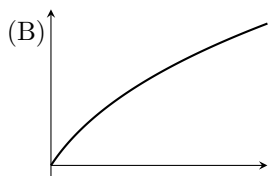
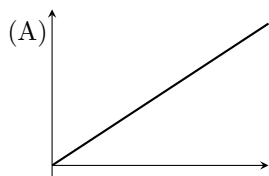
We then place the input variable next to the function name in parentheses, like $C(n)$. Note that in this case the parentheses do not denote multiplication. Instead, they are showing the reader what quantity is input into the function.

This is how we read and write this information: In the previous example, the sentence "Let $C(n)$ be the cost to buy n chairs," **defines the function**, meaning that it explains to the reader that for this problem, any time you see C or $C(n)$, we are referring to the function relationship where we input n , the number of chairs, and get the cost to buy n chairs as the output. For example, $C(10)$ is now a convenient way to refer to the cost to buy 10 chairs without doing any calculation at all.

3. A study has found that the shape of a glass affects how quickly people drink from it. This happens because people have trouble judging how the volume of liquid in a non-cylindrical glass relates to the height of the liquid.



Imagine you're pouring liquid into the glass shown to the right. Which of the following shows the height of the liquid as a function of the volume? Explain your choice.



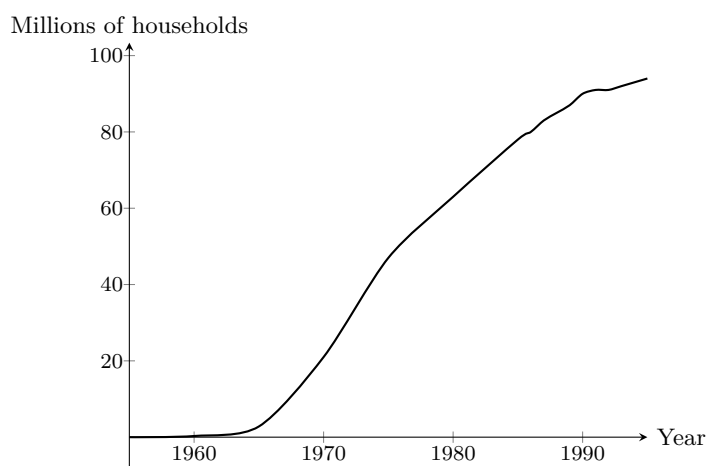
4. After a cup of hot chocolate is poured, the temperature cools off very rapidly at first, and then cools off more slowly, until the temperature of the hot chocolate eventually reaches room temperature. Let $H(t)$ be the hot chocolate's temperature in $^{\circ}\text{F}$ t minutes after it's been poured.

a) Sketch a graph of $H(t)$.

b) Is H increasing, decreasing, or constant? Is H concave up, concave down, or linear?

5. According to a New York Times article on the level of carbon dioxide in the atmosphere, “not only is the carbon dioxide level rising relentlessly, but the pace of that rise is accelerating over time.” What does this tell you about the graph of carbon dioxide level as a function of time?

6. Here’s a graph showing the number of millions of US households with a color television as a function of time.



Based on this graph, narrate how color TVs were adopted in the US. (How did the rate at which households adopted color TVs change over time? When was it speeding up? When was it slowing down? Can you come up with some plausible reason for that?)

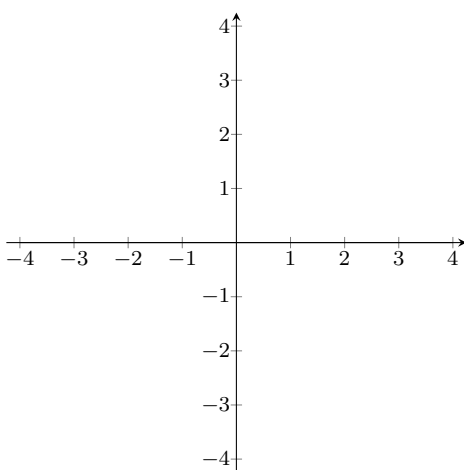
1.2 A LIBRARY OF BASIC FUNCTIONS AND PROPERTIES

You should be able to:

- Sketch and recognize graphs of $y = mx + b$, $y = x^2$, $y = x^3$, $y = \frac{1}{x}$, $y = \frac{1}{x^2}$, $y = |x|$, and $y = \sqrt{x}$.
- Understand the following characteristics of a function graphically: positive / negative, increasing / decreasing, intercepts, asymptotes, continuity, concavity, even / odd.
- Algebraically determine whether a function is even, odd, or neither.
- Explain what it means to say a function is increasing / decreasing *without bound*, as opposed to having an *upper bound* or *lower bound*.
- Recognize when the graph of a function given algebraically has a hole and construct algebraic examples of functions whose graphs have holes.

1. **Warm up.** Last time, we talked about what it meant graphically for a function to be concave up or concave down. Write down / sketch what these terms mean. Can a function be neither?

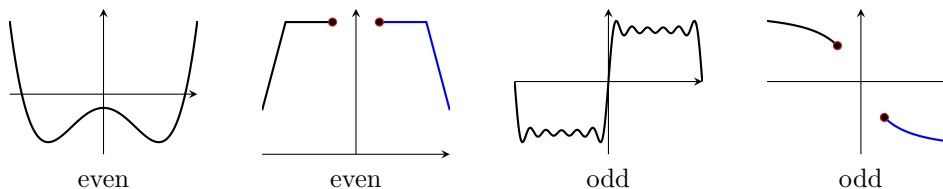
2. On the axes below, sketch graphs of $f(x) = 1$, $g(x) = x$, $h(x) = x^2$, and $j(x) = x^3$ (and label which is which).



3. *Characterizing functions.*

- a) Which of the functions in #2 are **positive** on $(-\infty, \infty)$?
- b) Which of the functions in #2 are **increasing** on $(-\infty, \infty)$?
- c) Which of the functions in #2 are **linear** on $(-\infty, \infty)$?
- d) Which of the functions in #2 are **concave up** on $(-\infty, \infty)$?

Two types of symmetry. We say that a function is even if its graph is symmetric over the vertical axis and odd if, when you rotate it 180° about the origin, the result is the same as the original. Here are a few examples:



- e) Which of the functions in #2 are **even**?

Which are **odd**?

Come up with algebraic definitions of even and odd:

- A function $f(x)$ is even if _____.
- A function $f(x)$ is odd if _____.

One more important property! The functions in #2 are all **continuous** on $(-\infty, \infty)$, meaning that we can sketch each graph without picking up our pencil.

4. For each of the following functions:

- What is the domain?
- Use the algebraic definitions of even and odd to determine whether the function is even, odd, or neither.
- Sketch the graph of the function.
- Where is the function concave up? Where is it concave down?

a) $f(x) = \frac{1}{x}$

b) $f(x) = \frac{1}{x^2}$

c) $f(x) = \sqrt{x}$

5. Let $f(x) = \frac{x^3 - x^2}{x - 1}$.

- a) What's the domain of f ? Use interval notation. How does the formula inform our understanding of the domain?
- b) In this problem, we'll sketch the graph of f . Before diving straight in, start by rewriting $f(x)$ by factoring. :

$$f(x) = \frac{x^3 - x^2}{x - 1} = \underline{\hspace{2cm}}$$

What does this re-written format tell us about the function $f(x)$?

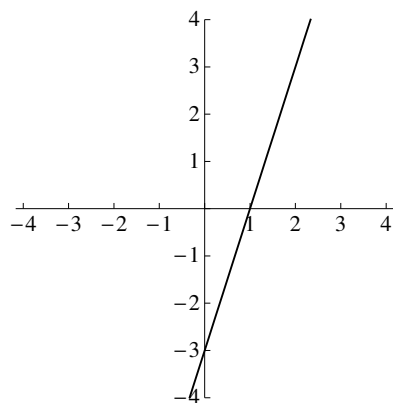
- c) Sketch the graph of f . Make sure your sketch is consistent with the previous parts of the problem.

- d) Is f continuous on $(-\infty, \infty)$?

e) Find a formula for $g(x)$, which is identical to x^3 , except $x = 0$ is not in the domain of the function.

6. In each part, think of a function $f(x) = \underline{\hspace{2cm}}$ whose graph looks like the given picture. Also, fill in the blanks / answer the other questions in each part.

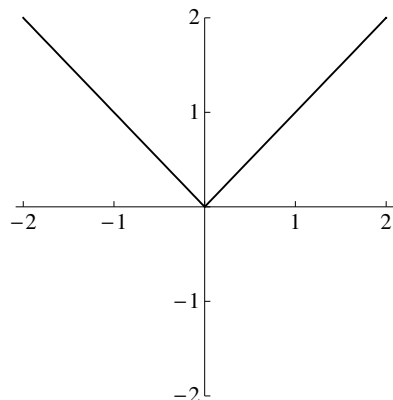
a)



- $f(x) = \underline{\hspace{2cm}}$
- f has x -intercept(s) $\underline{\hspace{2cm}}$
- f has y -intercept $\underline{\hspace{2cm}}$

Is f even? Is f odd? Explain both geometrically and algebraically.

b)



- $f(x) = \underline{\hspace{2cm}}$
- f is positive on $\underline{\hspace{2cm}}$
 f is negative on $\underline{\hspace{2cm}}$
- f is increasing on $\underline{\hspace{2cm}}$
 f is decreasing on $\underline{\hspace{2cm}}$
- The domain of f is $\underline{\hspace{2cm}}$
 The range of f is $\underline{\hspace{2cm}}$

We can see that the graph is made up of two pieces of lines. (So, we say f is *piecewise linear*.)
 What are the equations for those lines?

7. Reviewing the functions we've looked at so far, answer this true/false.

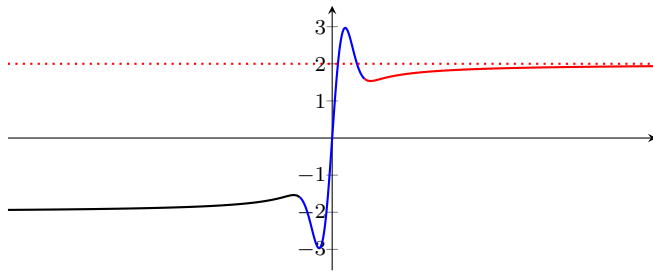
True or false: If a function has a formula that looks like $f(x) = \frac{\text{something}}{x}$, then f must have a vertical asymptote at $x = 0$.

8. a) Two of the functions in #4 are decreasing on $(0, \infty)$. Do they have a **lower bound** (a value that they never get below) on $(0, \infty)$, or do they **decrease without bound** on $(0, \infty)$?

- b) The other function in #4 is increasing on $(0, \infty)$. Does it have an **upper bound** (a value that it never gets above) on $(0, \infty)$, or does it **increase without bound** on $(0, \infty)$?

9. *A more interesting example of horizontal asymptotes.*

Here's the graph of a function $f(x)$ that has *two* horizontal asymptotes, which we've labeled on the graph. What are they (roughly)?



10. Sketch the graph of the function

$$f(x) = \begin{cases} 1 & \text{if } x < 3 \\ x & \text{if } x \geq 3 \end{cases}$$

11. Algebraically determine whether each of the following is even, odd, or neither.

a) $f(x) = -3x^4 + 6x^2 + 1$

b) $g(x) = x^3 + x^2 + 1$

1.3 MORE ON FUNCTIONS, AVERAGE RATES OF CHANGE

You should be able to:

- Write and interpret expressions in function notation.
- Compute an average rate of change, interpret it in the context of a situation, and represent it graphically.
- Explain the difference between average speed, average velocity, instantaneous speed, and instantaneous velocity.
- Explain how concavity can be characterized in terms of secant line segments.

1. *Warm up.* Your friend is preparing for a marathon and goes for a training run. What information would you need to be able to calculate your friend's average speed on this run?

2. In Problem Set 2, #6, you looked at a road trip east on Route 66 from Flagstaff, Arizona through New Mexico into Oklahoma. The road trippers entered Gallup, New Mexico T hours into the trip.

Let $f(t)$ be the number of miles they travel in the first t hours of their trip, and let $g(t)$ be their speed t hours into the trip.

a) How are these two functions related to instruments on the car's dashboard?

b) Express the following quantities in terms of f or g , whichever is appropriate.

i. The car's speed exactly 2 hours before entering Gallup.

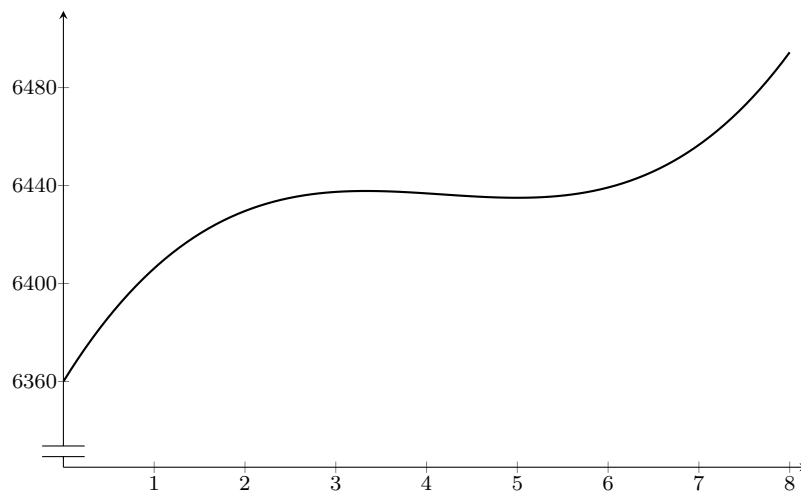
ii. The car's average speed between hour 2 and hour 5 of the trip.

iii. The car's average speed in the half hour after getting to Gallup, New Mexico.

c) In this situation, what do $g(T + 2)$ and $g(T) + 2$ represent? (Are they the same thing?)

d) Interpret $\frac{f(T+5) - f(T)}{5}$ in words. (What does it express about the road trip?)

3. Based on U.S. Census data, the population of Massachusetts between 2000 and 2008 can be modeled by the function $P(t) = 6360 + 60t - 15t^2 + 1.2t^3$, where $P(t)$ gives the number of thousands of people in Massachusetts t years after July 1, 2000.



- a) According to this model, what was the population of Massachusetts on July 1, 2000?

Copy the graph onto the board, and draw in your answers to the next few parts.

- b) Write an expression giving the change in population from July 1, 2001 to July 1, 2003. How could you visualize this quantity on the graph above?
- c) Write an expression giving the average rate of change of the population from July 1, 2001 to July 1, 2003 (with respect to time). What are the units of your answer? How could you visualize this quantity on the graph above?

- d) Put the following quantities in increasing order. (All 3 rates are with respect to time.)
- A. The average rate of change of the population between July 1, 2001 and July 1, 2002.
 - B. The average rate of change of the population between July 1, 2001 and July 1, 2003.
 - C. The average rate of change of the population between July 1, 2005 and July 1, 2006.

Average Rate of Change

The average rate of change of a function $f(x)$ over an interval is calculated by dividing the change in the output quantity's value by the change in the input quantity's value. If we define the output variable to be y and the input variable to be x , this calculation is $\frac{\Delta y}{\Delta x}$. You may recognize this as a slope calculation, even though our function $f(x)$ may not be a linear function.

The average rate of change tells us the constant rate of change necessary to achieve the same change in output over the same change in input. If we are calculating average rate of change over the interval $[a, b]$, we would do this by using:

$$\frac{f(b) - f(a)}{b - a}$$

Notice that the numerator is computing the change in the output quantity of the function f over the interval and dividing by the denominator, which is the change in the input quantity. Our unit of measurement for this rate are units of the output quantity per unit of the input quantity.

4. So far, we've described concavity only qualitatively. We can give a more precise definition using secant lines. Draw a few examples of functions that are concave up on an interval $[a, b]$, as well as a few examples of functions that are concave down on an interval $[a, b]$.

Use your graphs to fill in the following blanks with either "above" or "below".

- When a curve is concave up on an interval, its secant line segments in that interval lie _____ the curve.
 - When a curve is concave down on an interval, its secant line segments in that interval lie _____ the curve.
5. Roughly where is the graph in #3 concave up? Roughly where is it concave down?

Position versus distance, average speed versus average velocity

Position and distance traveled are common quantities that are used in many functions and models we'll build, so it's worth discussing the difference between the two. **Position** is a particular location, described relative to a starting point. For this reason, it includes a sign (positive or negative). For example, if a variable represents the position of an object north of a stop sign (in feet), a positive value would mean the object is located north of the stop sign and a negative value means the object is located south of the stop sign. **Distance traveled** typically has no sign attached to it. It is a positive number that represents how far an object has moved over the course of some time interval.

Speed and velocity have a similar relationship, where velocity has an orientation (meaning it can be positive or negative), and speed does not. When working with velocity, the problem description will include information that indicates which direction is positive. To calculate **average speed** and **average velocity**, we consider different calculations:

$$\text{average speed} = \frac{\text{distance traveled}}{\text{time elapsed}}$$

$$\text{average velocity} = \frac{\text{change in position}}{\text{time elapsed}}$$

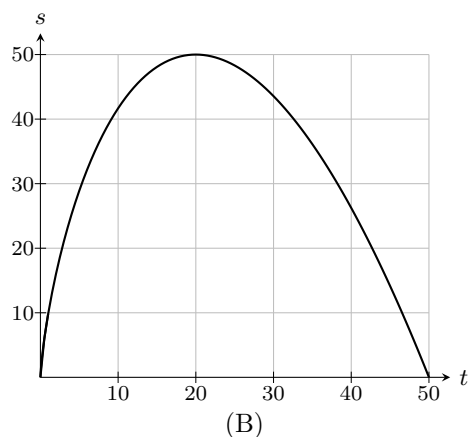
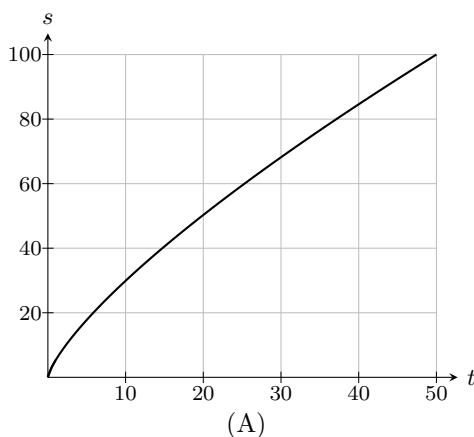
Example: Consider the situation where we run 0.5 miles due north and then turn around and run back to our starting point in the course of 15 minutes. These two calculations will offer different results. Our total distance traveled is 1 mile (since we ran half a mile there and back). Our change in position is 0 miles, since we ended where we started.

$$\text{average speed} = \frac{1 \text{ mile}}{0.25 \text{ hours}} = 4 \text{ mph}$$

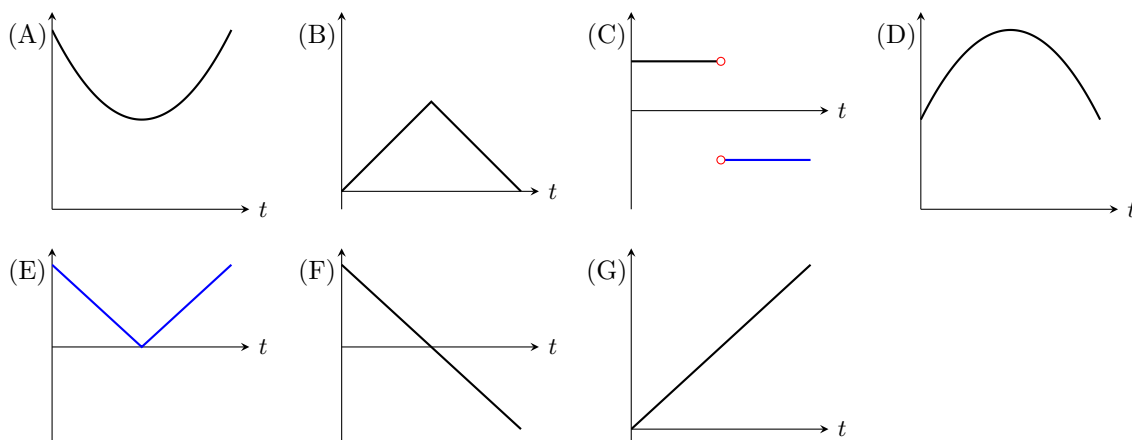
$$\text{average velocity} = \frac{0 \text{ miles}}{0.25 \text{ hours}} = 0 \text{ mph}$$

6. *Average speed vs. average velocity.* A swimmer is swimming a 100 m long race, which is one lap in a 50 m long pool. Let $s(t)$ be the swimmer's distance from the starting position t seconds after the start of the race.

- a) Which of the following is a more reasonable graph for $s(t)$? Why?



- b) According to the graph you chose, what was the swimmer's average speed for the race? Average velocity for the race?
- c) What was the swimmer's average speed over the last 50 m of the race? Average velocity?
- d) What was the swimmer's average speed over the first 20 seconds of the race? Average velocity?
- e) What's the difference between velocity and speed?
- f) Let $v(t)$ be the swimmer's velocity at time t . Interpret $\frac{v(15) - v(10)}{5}$ in words. What are its units?
7. From your height of 1.5 m, you throw a ball straight up into the air. Pick the graph (choices A - G below) that could be the ball's...
- a) height as a function of time (for the first few moments after you toss the ball)
- b) velocity as a function of time
- c) speed as a function of time



8. In this problem, we'll look at the function $f(x) = x^2$. Calculate the average rate of change of f on the following intervals.

a) $[1, 3]$

b) $[1, 1.5]$

c) $[1, 1 + h]$. Fully simplify by expanding and then combining like terms. (From your work in (a) and (b), you should already know the answer for certain values of h ; which ones? Does your answer here agree with your answers to (a) and (b)?)

d) Sketch the graph of $f(x) = x^2$, and draw secant lines whose slopes correspond to the average rates of change in (a) and (b). Which of the average rates of change is larger?

9. Suppose $f(x)$ is a linear function defined on $(-\infty, \infty)$. How does the average rate of change of f on $[0, 5]$ compare to the average rate of change of f on $[0, 500]$? What about the average rate of change of f on $[-50, 2]$?

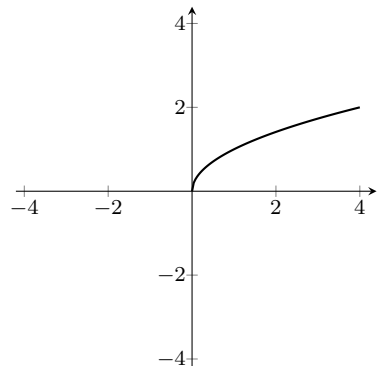
1.4 TRANSFORMING FUNCTIONS

You should be able to:

- Sketch $y = af(bx) + c$, $y = af(x + b) + c$, and $y = |f(x)|$ from a graph of $y = f(x)$.

1. In each part, the graph of $f(x) = \sqrt{x}$ is shown, and you're given a new function. Write a formula for the new function and sketch its graph. What are the domain and range of the new function?

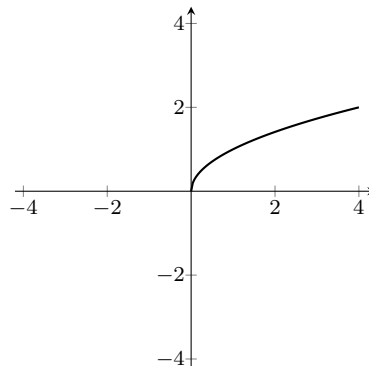
a) $a(x) = -f(x) =$ _____



Domain of a : _____

Range of a : _____

b) $b(x) = f(-x) =$ _____

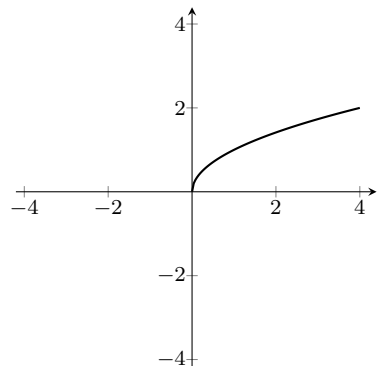


Domain of b : _____

Range of b : _____

Again, you're given a new function based on $f(x) = \sqrt{x}$. If you want to graph the new function, what type of transformation should you apply to $y = \sqrt{x}$? What's one piece of information (like the domain, range, x -intercepts, or y -intercept) you could use to check which direction the transformation goes? Sketch the graph on the given axes.

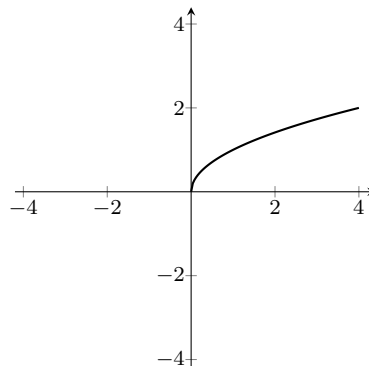
c) $c(x) = f(x) + 2 =$ _____



Type of transformation: _____

Useful info: _____

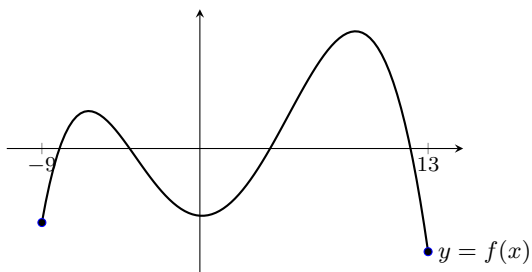
d) $d(x) = f(x + 2) =$ _____



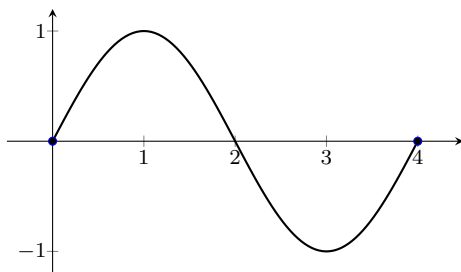
Type of transformation: _____

Useful info: _____

2. Here's the graph of a function f with domain $[-9, 13]$. On the same axes, sketch $y = |f(x)|$.



3. Here's the graph of a function f .



- a) What are the domain, range, and intercepts of f ? (Give both the x - and y -intercepts.)

In each part, you're given a function. Sketch the graph of the function, identify its domain and range, and say as much about the intercepts as you can.

b) $-2f(x)$

d) $-3f(x + 1)$

c) $f(2x)$

e) $3f(2x) - 4$

4. **Transformations in function formulas.** In each part below, you're given a function whose formula is the result of transforming $y = \frac{1}{x^2}$. (Part a) is the original function itself, which offers us our starting point. Identify what transformations have occurred and sketch the graph of the function, labeling any asymptotes. For (b) - (e), explain how the graph is related to the graph of $y = \frac{1}{x^2}$.

a) $y = \frac{1}{x^2}$

d) $y = \frac{1}{(x+4)^2} + 2$

b) $y = \frac{3}{x^2} - 1$

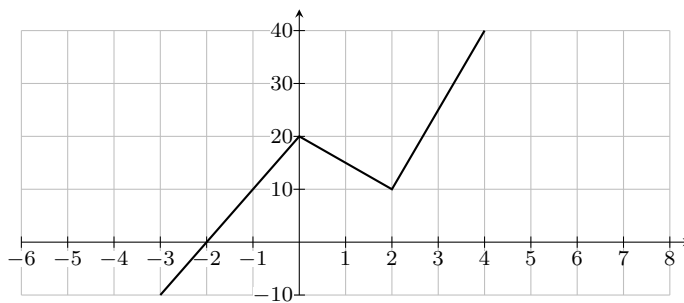
e) $y = -\frac{3}{(x-2)^2} - 1$

c) $y = 3\left(\frac{1}{x^2} - 1\right)$

5. **Revisiting the algebraic definitions of even and odd functions.** We've seen that a function f is even if $f(-x) = \underline{\hspace{2cm}}$ and odd if $f(-x) = \underline{\hspace{2cm}}$.

Now, we can understand this in terms of graph transformations. How does the graph of $f(-x)$ relate to the graph of $f(x)$? Why does this make sense with the definitions of even and odd?

6. Here's the graph of a function f which has domain $[-3, 4]$:



Let $g(x) = f(2x)$.

- a) Is $g(3)$ defined or undefined?

- b) What's the domain of g ?

- c) Fill in the blanks:

$$g(\underline{\hspace{1cm}}) = f(-3) = -10$$

$$g(\underline{\hspace{1cm}}) = f(0) = 20$$

$$g(\underline{\hspace{1cm}}) = f(2) = 10$$

$$g(\underline{\hspace{1cm}}) = f(4) = 40$$

Use this to graph g on the axes above.

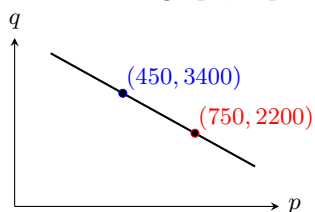
1.5 LINEAR AND PIECEWISE LINEAR FUNCTIONS

You should be able to:

- Write the equation of a line given 2 points or 1 point and the slope, and decide whether slope-intercept form ($y = mx + b$) or point-slope form is more useful.
- Use linear and piecewise linear equations to model various phenomena.
- Interpret the slope in linear and piecewise linear functions, and graph the “slope function” $f'(x)$ of a piecewise linear function $f(x)$.

1. Economists use demand curves to express the relationship between the price of an item and the number of items customers want to buy (the “demand”). Below is the demand curve for a certain good. p is the price per item measured in dollars, and q is the quantity of the good demanded (in other words, the number of items customers want to buy).

- a) Based on the graph, express the demand q as a function of the price p .



- b) In words, explain the meaning of the slope of the line.

- c) If the item is priced at \$725, how many items will customers want to buy?

2. Find an equation for each of the following lines.
- a) The line with slope $-\frac{1}{5}$ passing through the point $(23, 71)$.

b) The line passing through the points $(0, p)$ and $(q, 0)$.

3. In June 2023, the cost to send a letter through the US Postal Service was:

Weight Not Over (oz.)	
1	\$0.63
2	0.87
3	1.11
3.5	1.35

Let $C(w)$ be the cost to mail a letter weighing w ounces.

- a) Sketch the graph of $C(w)$. What is its domain?

b) Is $C(w)$ continuous on its domain?

c) Write a formula for $C(w)$.

4. Molly has just started a part time job that pays \$10 per hour for the first 6 hours that she works each week, and \$12 per hour for each hour after that. Molly is allowed to work up to 15 hours per week. Let $P(t)$ be Molly's income in a week if she works t hours.

a) Find a formula for $P(t)$.

b) Sketch the graph of $P(t)$.

c) Graph the “**slope function**” $P'(t)$, which still takes t , time worked, as input, but gives the slope of the function $P(t)$ as its output.

d) What is the practical meaning of $P'(t)$? Include units in your answer.

5. Instead of a straight royalty percentage on sales of his book, *The Story of a Novel*, the author Thomas Wolfe agreed to accept a sliding scale of 10% on the first 3000 copies sold, 20% on the next 1500 copies, and 30% on all copies after that. The book was priced at \$2.⁽¹⁾

a) Write a function $R(x)$ that gives Wolfe's royalties as a function of x , the number of books sold.

b) Graph $R(x)$.

c) Do some sort of check of whether your answer to (a) is sensible, as well as whether it agrees with your graph in (b). (How you decide to check is up to you; thinking of a good way to check is part of the challenge!)

⁽¹⁾The numbers in this problem have been modified for computational ease.

d) Solve $R(x) = 900$. What does this equation mean?

e) Graph the “slope function” $R'(x)$, which gives the slope at every point on the graph of R .

f) What’s the practical meaning of $R'(x)$? Include units in your answer.

1.6 FUNCTION COMPOSITION, MODELING WITH FUNCTIONS

You should be able to:

- Explain how composition of functions models situations where there is a chain of dependency among variables.
- Compute compositions of functions.
- Model situations systematically: identify your goal, define variables and express your goal in terms of those variables, and then look for relationships among your variables.
- Communicate your modeling process clearly.

1. The price of tomatoes varies seasonally; suppose $P(x)$ gives the price of tomatoes in dollars per pound at time x . The demand for tomatoes (in other words, the amount of tomatoes people want to buy) depends on the price; say that $D(x)$ gives the demand for tomatoes when they cost $\$x$ per pound.

Answer the following questions using functional notation.

a) When tomatoes cost $\$3$ per pound, what is the demand for tomatoes?

b) At time 3, what is the demand for tomatoes?

c) What's the demand for tomatoes at time t ?

2. **Working algebraically with compositions.** If $f(x) = x^2$ and $h(x) = 2x - 1$, find each of the following.

a) $f(h(x))$

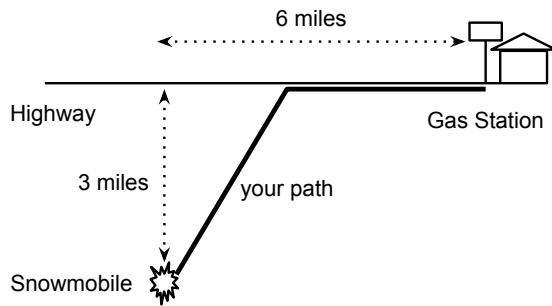
b) $h(f(x))$

3. a) Harvard's Hollis Professor of Divinity (currently Karen King) has the right to graze animals in Harvard Yard.⁽²⁾ Professor King wants to have some room for her cow to roam in without escaping, so she decides to build a rectangular fenced-in pen using 28 feet of wire fencing. She'd like to understand how much area the pen could have, so she asks you to model the area of the pen as a function of the length of one side. Help her out! (Assume that she will use all 28 feet of fencing.)

- b) What's the domain of the function in (a)?

⁽²⁾<https://news.harvard.edu/gazette/story/2012/05/the-oldest-endowed-professorship/>

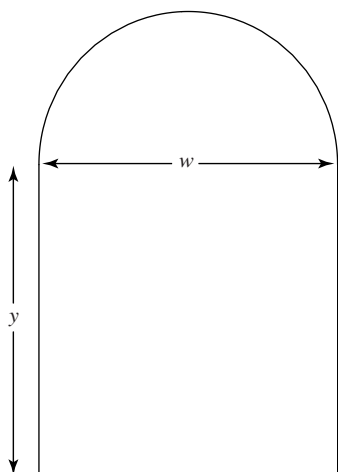
4. Your snowmobile runs out of gas when you're 3 miles south of a major highway. The nearest gas station is on the highway 6 miles east of your position. You decide to head for some point on the highway and then walk along the highway until you reach the gas station, as shown in the diagram.



- a) If you choose a path that involves walking h miles along the highway, how many miles will you walk through the snow? (Express your answer in terms of h .)
- b) You can walk 3 miles per hour on roads but only 2 miles per hour through snowy fields.
If you choose a path that involves walking h miles along the highway, how long will it take you to reach the gas station? (Again, express your answer in terms of h .)

5. A new mosque is being built in the Turkish town of Iznik. Under construction is an archway whose structure will be a rectangle capped by a semicircle, as shown in the figure below. The distance from the highest point of the arch to the floor is 7 meters. Denote the width of the archway by w and the height of the vertical wall of the rectangle y . The width and height are given in meters.

- a) Express y , the height of the side wall, as a function of w (meaning, in terms of w), the width of the archway.
- b) Express the area enclosed by the archway as a function of w .
- c) Express the perimeter of the archway (excluding the floor) as a function of y .



6. A cistern for storing water is in the shape of a right circular cone with radius 8 feet and height 40 feet.⁽³⁾ (The tip is at the bottom.)
- a) Suppose the cistern is filled part-way with water. Express the height of the water as a function of the radius of the surface of the water. What's the domain of this function?
 - b) Express the volume of water in the cistern as a function of the height of water in the cistern. What's the domain of this function?

⁽³⁾Useful fact: The volume of a cone with radius R and height H is $\frac{1}{3}\pi R^2 H$.

1.7 APPROXIMATING WITH SECANT LINES

You should be able to:

- Understand what local linearity is and why it is useful for determining which approximation methods are good options when modeling how a particular quantity is changing.
- Decide where the graph of a function is or isn't locally linear.
- Use secant lines to approximate function values.
- Determine whether a secant line approximation is too high or too low, or if there isn't enough information to be sure.

1. **Warm up!** From memory, try to recall the formula we use to compute the average rate of change of the function $f(x)$ over the interval $[a, b]$.

2. Here is some sunrise information for Cambridge, MA in 2023, from timeanddate.com:

	Sep 17	Sep 18	Sep 19	Sep 20
Sunrise	6:26 am	6:27 am	6:28 am	6:29 am

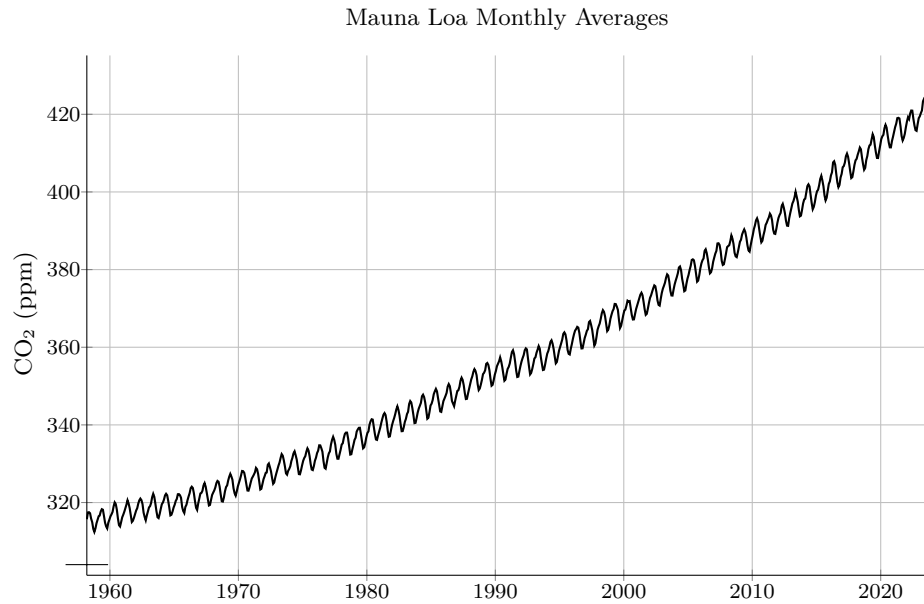
a) Estimate when the sun will rise on September 30.

b) What assumptions did you make to come up with your estimate in (a)?

c) Would you feel comfortable using the same assumptions to estimate the sunrise time on February 29, 2024? Why or why not?

In #1, we introduced the concept of **local linearity**. In the space below, write the definition of local linearity in your own words:

3. At the Mauna Loa Observatory in Hawaii, the amount of carbon dioxide in the atmosphere has been measured every month since 1958.⁽⁴⁾ Here's a graph of the data.

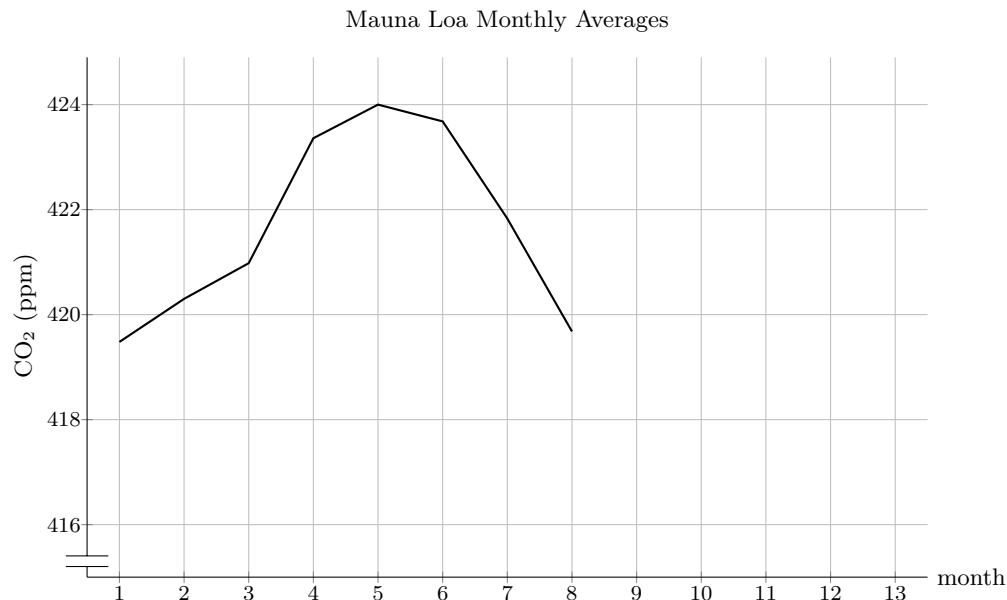


Generally, the carbon dioxide concentration each year is highest in May and lowest in October.

- a) A scientist is looking at some recent measurements, from June 2023 (423.68 ppm) and August 2023 (419.68 ppm). Do you think it's reasonable for her to use local linearity to predict the CO_2 concentration in September 2023? If so, find the predicted value for her; if not, explain why not.

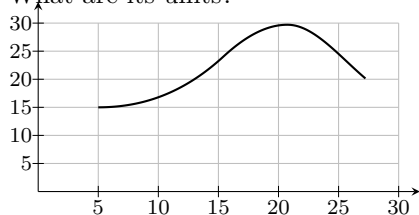
⁽⁴⁾Because Mauna Loa Volcano erupted in November 2022, readings after that are actually from Maunakea Observatories, which is about 21 miles away from the Mauna Loa Observatory.

- b) Here's a graph showing just the measurements from 2023. (January 2023 = 1, February 2023 = 2, and so on.) Illustrate the estimate you made in (a).



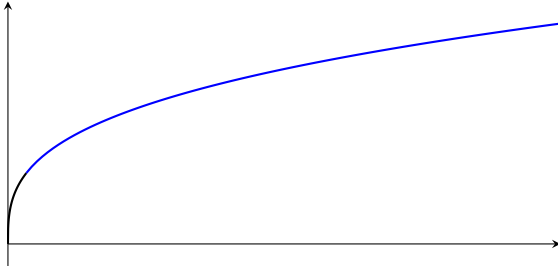
- c) In (a), you used a line to make an approximation. Explain what the slope of that line means; include units in your answer.
- d) Is it reasonable to use the line from (a) to predict the CO₂ concentration in January 2024? If so, find the predicted value; if not, explain why not.

4. The maximum swimming speed of salmon depends on the water temperature. Suppose $S(T)$ is the maximum swimming speed (in cm/s) of salmon when the water temperature is T degrees Celsius; the graph of S is shown below. What does the average rate of change of S on the interval $[15, 20]$ represent? What are its units?



5. a) *Reminder:* $\sqrt[3]{x}$ can also be written as x raised to what power?

Here's the graph of $f(x) = \sqrt[3]{x}$ on $[0, 30]$.



- b) Use an appropriate secant line to approximate $\sqrt[3]{10}$ by hand. Illustrate your approximation on the graph.

- c) Is your estimate an overestimate (too high) or an underestimate (too low)? How can you be sure?

Another way we can say this is that a(n) _____ bound for $\sqrt[3]{10}$ is _____.

- d) Use the secant line that you found in (b) to estimate $\sqrt[3]{7.5}$. Is your estimate too high or too low?

In other words, a(n) _____ bound for $\sqrt[3]{7.5}$ is _____.

- e) *Reflect.* What form of the equation for a line was most useful in this problem?

Our calculations in the previous problem were over an interval of a specific length. In this problem, we'll explore how to find the average rate of change over an interval of variable length using the increment variable h .

6. Let $f(x) = x^2 - x$.

a) Find the average rate of change of $f(x)$ between $x = 3$ and $3 + h$, where $h \neq 0$. Your final answer will be an expression that will include 3 and h .

i. Use the expression you found in (a) to calculate the average rate of change of $f(x)$ over the interval $[3, 3.25]$

ii. Evaluate the average rate of change of $f(x)$ over the interval $[3, 3.05]$.

b) Can you use the expression to determine the average rate of change of $f(x)$ over the interval $[2.9, 3]$ using the expression in (a)? If it's possible, do it! If it's not possible, explain why not.

7. A rocket is launched straight up into the air. Suppose the function $f(t) = t^2$ describes the rocket's height in meters t seconds after liftoff, and we'd like to know the rocket's *instantaneous* velocity 5 seconds after liftoff. That is, we'd like to know how fast the rocket is going right at the instant $t = 5$.

a) We don't know how to find this instantaneous velocity exactly. Instead, come up with a good estimate for this velocity, and explain how you came up with your estimate.

b) How can you improve your estimate?

8. An ice cream shop is keeping track of how their sales are affected by price changes. When they sell a single scoop of ice cream for \$4, they sell 600 single scoops in a day. When they raise the price to \$4.50, they sell 560 single scoops in a day.

Let $S(p)$ be the number of single scoops they sell in a day when the price is \$ p .

a) What does the average rate of change of $S(p)$ on $[4, 4.5]$ represent? What are its units?

b) Do you think it's reasonable to use local linearity to approximate the number of scoops the ice cream shop would sell if they changed the price of a scoop to \$4.20? How about if they changed the price to \$10?

UNIT 2

THE DERIVATIVE

2.1 DEFINITION AND INTERPRETATION OF THE DERIVATIVE

You should be able to:

- State the definition of the derivative, explain where it comes from, and use it to calculate derivatives.
- Interpret the derivative in context and represent it graphically.
- Find the tangent line to the graph of $f(x)$ at $x = a$.
- Use tangent lines to approximate function values, and determine whether such an approximation is too high or too low, or if there isn't enough information to be sure.

1. A rocket is launched straight up into the air. Suppose the function $f(t) = t^2$ describes the rocket's height in meters t seconds after liftoff.
 - a) Sketch the graph of $f(t)$ for t between 0 and 7. Determine its average velocity over $[0, 7]$.

Suppose you'd like to know the rocket's *instantaneous* velocity 5 seconds after liftoff. That is, you'd like to know how fast the rocket is going right at the instant $t = 5$.

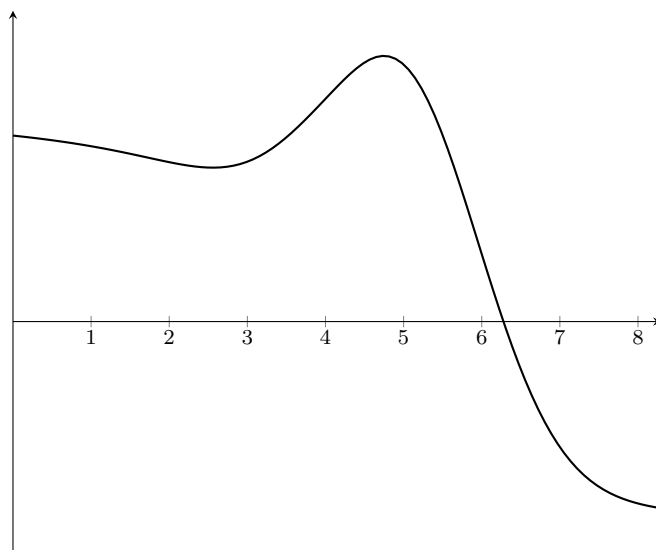
- b) Would it work to find the rocket's average velocity between $t = 5$ and $t = 5$? Why or why not?
- c) How can we *approximate* this instantaneous velocity? Write a general expression that *approximates* the instantaneous velocity. How would you visualize your expression graphically? Illustrate it on your graph.
- d) How could we find this instantaneous velocity *exactly*? How would you visualize this instantaneous velocity graphically? Illustrate it on the graph.

d) Calculate $f'(1)$ using the definition of the derivative. Is this consistent with your answer to (a)?

e) Find the equation of the line tangent to $y = \frac{1}{x}$ at $x = 1$.

3. *Practice interpreting notation.*

Here's the graph of a function f .



In each part, you're given an expression. Sketch a line whose slope is the given expression.

a) $\frac{f(6) - f(3)}{3}$

b) $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$

c) $\lim_{w \rightarrow 3} \frac{f(w) - f(3)}{w - 3}$

d) $\lim_{x \rightarrow 6} \frac{f(x) - f(4)}{x - 4}$

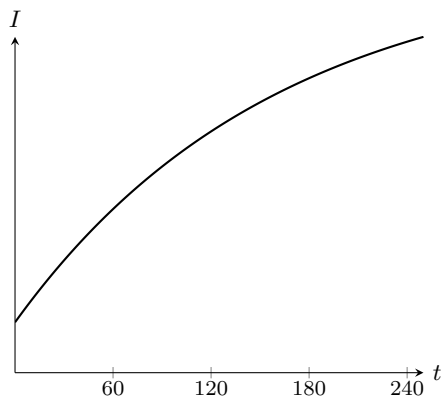
e) $\lim_{h \rightarrow 3} \frac{f(1+h) - f(1)}{h}$

4. Peter is roasting a 14 lb turkey as a test run for Thanksgiving. He starts roasting the turkey at noon. At 1:00pm, he checks on the turkey using a fancy thermometer; he discovers that it has an internal temperature of 35.6°C ,⁽¹⁾ and its temperature is rising at an instantaneous rate of 0.25°C per minute.

a) Approximate the temperature of the turkey at 1:06pm.

b) Let $I(t)$ be the turkey's internal temperature (in $^{\circ}\text{C}$) t minutes after noon. Use function notation to express what you were told about the turkey.

c) Here is a graph of $I(t)$. Use a sketch to explain the approximation you made in (a).



d) Based on your sketch, was your approximation too high or too low?

e) Would you be comfortable using the same method to predict the turkey's temperature at 3 pm? Why or why not?

⁽¹⁾According to the USDA, a turkey should be roasted to an internal temperature of 73°C .

2.2 THE DERIVATIVE FUNCTION

You should be able to:

- Use the limit definition to compute the derivative function f' .
- Determine whether a function is *differentiable*, both graphically and algebraically, and give examples of functions that are differentiable and non-differentiable.
- Sketch a rough graph of f' when you have the graph of f .

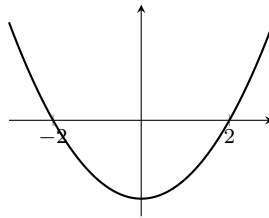
1. **Warm up!** Suppose f is a function.

a) State the definition of $f'(a)$. Explain how this definition effectively calculates the instantaneous rate of change.

b) How do we interpret $f'(a)$ graphically?

c) If $f(t)$ represents the position at time t of a runner along a running path, how could we interpret $f'(a)$?

2. Here's the graph of a function $f(t)$; we don't know a formula for $f(t)$, so answer the questions below using just the graph.



- a) i. Which is greater, $f'(\frac{1}{2})$ or $f'(1)$?
- ii. Which is greater, $f'(-3)$ or $f'(-1)$?
- iii. For what values of t will $f'(t)$ be positive?
- iv. For what values of t will $f'(t)$ be negative?
- v. Is f even, odd, or neither? Is f' even, odd, or neither? Explain.
- b) Sketch a rough graph of f' . Even though your sketch does not need to be perfect, it should reflect the information related to what we found above.

Note: Until this problem, we've been evaluating the derivative at a given point (for example, a). By reasoning about the derivative at various points in the domain of the function, we were able to sketch a picture of the more general **derivative function** $f'(t)$, which takes t as input, and gives the value of the derivative of f at t as output. Looking at our sketch of $f'(t)$, the y -value gives us the slope of the tangent line (the value of the derivative) on the original function at that input value.

- c) One of these is the formula for $f(t)$. Which one?

$$t^2 + 4 \quad t^2 - 4 \quad (t - 2)^2 \quad 4t^2$$

- d) Use the formula you picked in (c) to find a simplified formula for $f'(t)$. Does your result agree with your answers to (a)?

Notation. The function $f'(t)$ is also written as $\frac{df}{dt}$. If $f(t) = t^2$, as in this problem, we can also write $\frac{d}{dt}t^2$ instead of $f'(t)$. (We read $\frac{d}{dt}$ as “the derivative of”.)

3. a) Sketch the graph of $|x - 1|$.

- b) Sketch the graph of $\frac{d}{dx}|x - 1|$. (The notation $\frac{d}{dx}|x - 1|$ is shorthand for “the derivative of $|x - 1|$ ”.) There’s one particularly interesting value of x here; what’s going on at that value of x ?

Definition. We say a function f is non-differentiable at c if $f'(c)$ does not exist. Otherwise, we say that f is differentiable at c . (For both definitions, we only consider c that are in the domain of f .)

4. Gromit has been growing a giant squash for Tottington Hall's annual Giant Vegetable Competition. He has carefully tracked his squash's length and weight. Let $w(\ell)$ be the squash's mass in kg when its length is ℓ cm.

a) What are the units of $w'(\ell)$?

b) Another notation for $w'(\ell)$ is $\frac{dw}{d\ell}$. Do you expect $\frac{dw}{d\ell}$ to be positive or negative? Why?

c) Interpret the statement $w'(105) = 10$ in words.

d) Suppose $w'(70) = 8$. (This statement can also be written as $\left. \frac{dw}{d\ell} \right|_{\ell=70} = 8$.) Which is the following is the most reasonable conclusion?

- A. It takes 70 days for the squash to grow to be 8 kg.
- B. When the squash is 70 cm long, it weighs 8 kg.
- C. When the squash is 70 cm longer than its current length, it will weigh about 8 kg more than it currently does.
- D. When the squash grows from 70 cm to 71 cm, it will gain about 8 kg in mass.

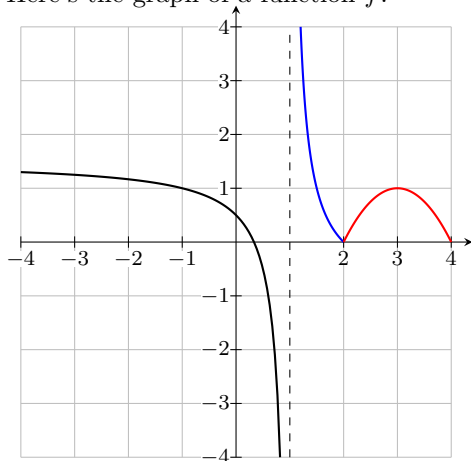
5. a) Sketch the graph of $f(x) = (x - 2)^3 + 1$.

b) Using the graph of f , sketch the graph of f' .

c) Sketch the graph of $g(x) = |(x - 2)^3 + 1|$.

d) Using the graph of g , sketch the graph of g' .

6. Here's the graph of a function f .



- For what value(s) of x in $(-4, 4)$ is f non-differentiable? How can you tell the function is non-differentiable there?
- For what value(s) of x in $(-4, 4)$ is $f'(x) = 0$? Visually, how can you tell if the derivative is 0 there?
- For what value(s) of x in $(-4, 4)$ is $f'(x) > 0$? How can you tell the derivative is positive there?
- Sketch the graph of f' on $(-4, 4)$.

2.3 FUNCTIONS AND THEIR DERIVATIVES

You should be able to:

- Explain how concavity can be characterized in terms of tangent lines.
- Explain what f' and f'' tell us about the graph of f . (For example, if f' is positive, what does that say about f ? If f' is increasing, what does that say about f ?)
- Explain how displacement, velocity, and acceleration are related in terms of derivatives.

1. **Warm up.** Sara, the owner of a cupcake truck, has found that the number of cupcakes she sells per day depends on the price she charges. Let $C(p)$ be the number of cupcakes she sells when the price of a cupcake is p cents. Suppose Sara has found that $C(300) = 550$ and $C'(300) = -2$. Using this information, which of the following can you find exactly? Which can you reasonably approximate?

a) The number of cupcakes Sara will sell tomorrow.

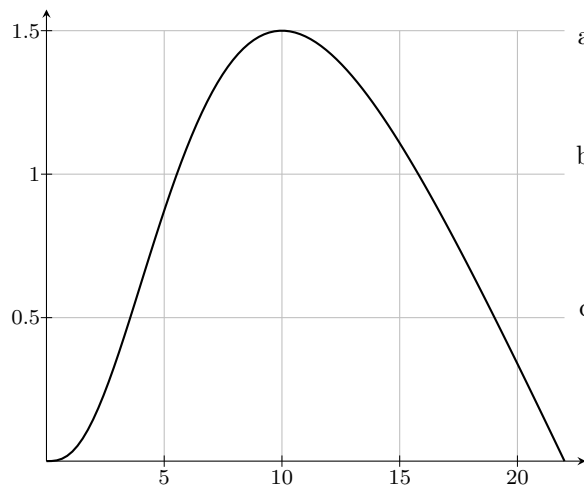
b) The number of cupcakes Sara will sell in a day if she charges \$5 per cupcake.

c) The number of cupcakes Sara will sell in a day if she charges \$2.95 per cupcake.

2. Previously, we described the relationship between concavity and secant lines. Now, let's think about tangent lines!

- When a curve is concave up, its tangent lines lie _____ the curve.
- When a curve is concave down, its tangent lines lie _____ the curve.

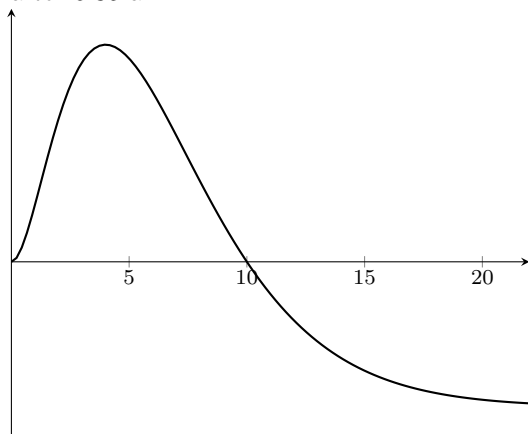
3. **Velocity and speed, average and instantaneous.** The #68 bus runs along a straight road between Harvard Square and Kendall Square. The first bus leaves Harvard Square at 6:35 am. Let $b(t)$ be the bus's distance in miles east of Harvard Square t minutes after 6:35 am. Here is the graph of $b(t)$:



- a) How many miles long is the bus route?
- b) At roughly what time does the bus reach Kendall Square?
- c) At roughly what time does it return to Harvard Square?
- d) Sketch a graph showing the bus's velocity t minutes after 6:35 am and another showing the bus's speed t minutes after 6:35 am. Begin by stating the relation between $b(t)$ and velocity.

- e) What is the bus's average velocity over this round trip?
- f) What is the bus's average speed over this round trip?
- g) What is the relationship between the bus's acceleration and its velocity? At roughly what times does the bus have positive acceleration? At roughly what times does it have negative acceleration?

- h) The picture below is the actual graph of $v(t)$, the bus's velocity t minutes after 6:35 am. (Does your answer to (d) match this qualitatively?) Sketch a graph of the bus's acceleration t minutes after 6:35 am.



- i) What is the relationship between the bus's acceleration and its distance in miles from Harvard Square?

Summarize your observations in the table below. (If there is no useful property, just leave the space blank.)

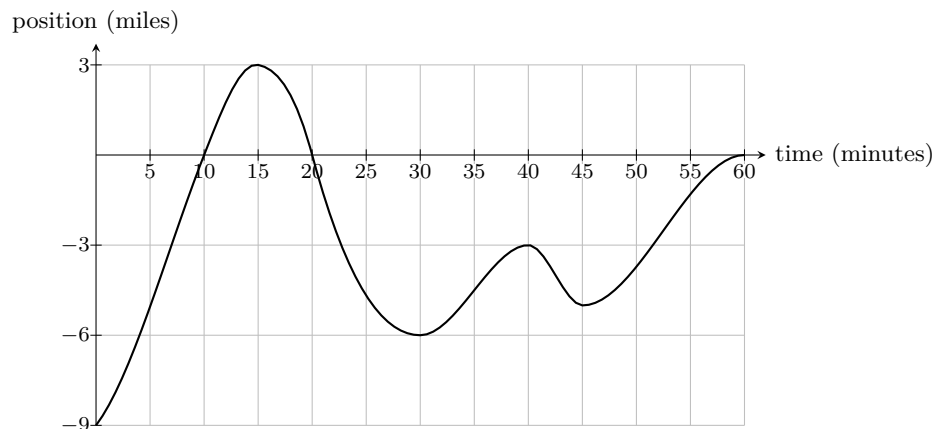
f	f'	
positive		
negative		
increasing ↗		
decreasing ↘		
concave up ∪		
concave down ∧		

4. Seulgi is trying to memorize a set of vocabulary words for her French class. Let $W(t)$ be the number of words she knows after t hours of study. She learns new words most quickly at first; as time goes on, her rate of learning keeps getting slower.

- What does the last sentence tell you about $W'(t)$?
- What does the last sentence tell you about $W''(t)$?
- What does the last sentence tell you about $W(t)$?

5. North Dakota's Highway 46 is supposedly the straightest road in the United States. From the city of Streeter, it runs 123 miles east to Hickson. Litchville is a tiny city on this highway.

A group of friends decides to drive on the highway one day. Let $s(t)$ be their position at time t , where t is measured in minutes after noon and the position is given in miles east of Litchville. Here is the graph of $s(t)$:

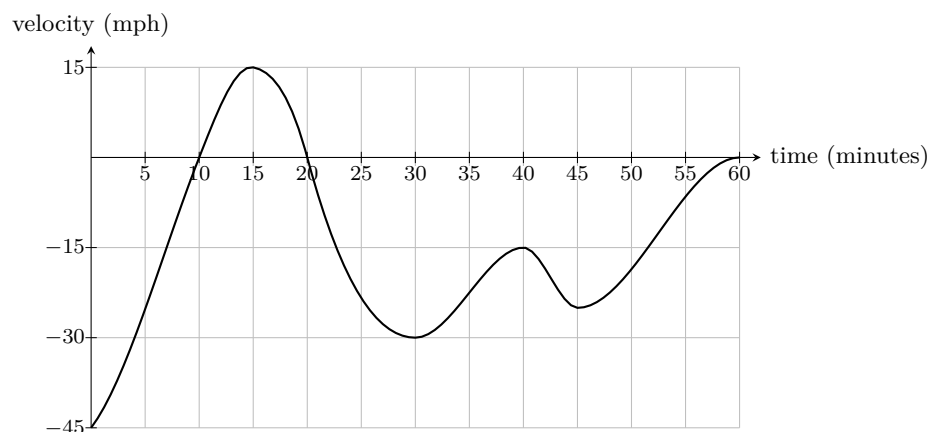


- When are the friends traveling east?

What did you look at to decide?

- Are the friends going faster at 12:05 or at 12:35? How do you know?

Now, suppose that a different group of friends is driving on the same highway, and that this is the graph of their **velocity** at time t :



c) When are the friends traveling east?

What did you look at to decide?

d) Are the friends going faster at 12:30 or 12:35? How do you know?

6. Let $f(x) = (x - 2)^3 + 1$.

a) Sketch a graph of f .

b) Based on your graph of f , sketch a graph of f' .

c) Use the definition of the derivative to calculate $f'(2)$.

d) How is the value of $f'(2)$ reflected in your graph of f ? How is it reflected in your graph of f' ?

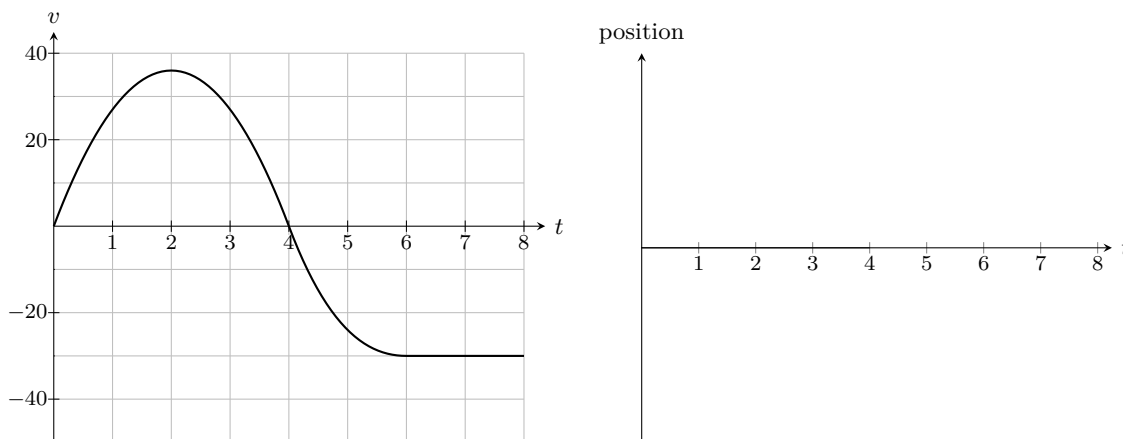
2.4 USING DERIVATIVES TO GRAPH FUNCTIONS

You should be able to:

- Use the first derivative to determine where a function is increasing or decreasing and to find *turning points*.
- Use the second derivative to determine where a function is concave up or concave down and to find *inflection points*.
- Put the above information together to sketch a reasonable graph of a function.
- Develop the habit of checking for consistency when putting several pieces of information together.

1. For a function $f(x)$, summarize the relationships among features (positive / negative, increasing / decreasing, etc.) of f , f' , and f'' .

2. Albert is taking a very long car trip. Suppose that the following graph gives his velocity v (measured in mph) as a function of time t (measured in hours).



Let $A(t)$ be Albert's position at time t .

- a) Where is $A(t)$ increasing? Where is $A(t)$ decreasing? How can you tell?
 - b) Where is $A(t)$ concave up? Concave down? Are there places where it's neither? How can you tell?
 - c) Suppose that, at time 0, Albert is at position 0. On the axes above, sketch a graph of $A(t)$. The main thing to focus on is making your graph consistent with your answers to (a) and (b).
 - d) Bart is also driving on the highway; at time 0, he's at position 10. By some miracle, his velocity graph is exactly the same as Albert's. Let $B(t)$ be Bart's position at time t , and sketch a graph of $B(t)$ on your graph from (c).
 - e) How does $B(t)$ relate to $A(t)$?
3. Generalize: If $f'(t)$ is continuous and $f'(t) = g'(t)$, how does $g(t)$ relate to $f(t)$?

4. In Problem Set 10, #3(a), you found that $\frac{d}{dx}(ax^2 + bx + c) = 2ax + b$. That will be useful in this problem!

Suppose f is an unknown function; all we know is that f is continuous on $(-\infty, \infty)$ and $f'(x) = x^2 + x - 6$. ($x^2 + x - 6$ is also continuous on $(-\infty, \infty)$.)

- a) Where (for what values of x) is $f'(x)$ positive? Where is $f'(x)$ negative? What does this tell you about f ?

- b) Find f'' and make a **sign chart** for f'' . What does this tell you about f ?

- c) **Important definition:** Values of x in the domain of f where $f(x)$ switches from increasing to decreasing or vice versa are called turning points. What are the turning points of f ?

- d) **Important definition:** Values of x in the domain of f where $f(x)$ switches from concave up to concave down or vice versa are called inflection points. What are the inflection points of f ?

- e) Sketch a possible graph of f (that's consistent with your answers so far).
- f) How many zeros (x -intercepts) could f possibly have? Sketch a possible graph for each possible number of zeros. How do the different possibilities relate to each other?
5. Suppose f is an unknown function with domain $(0, \infty)$; we know $f(x)$ has a vertical asymptote at $x = 0$ and that $f'(x) = x - \frac{4}{x}$.
- a) Where is f increasing? Where is f decreasing? What are the turning points of f ?

b) Use the definition of the derivative to calculate $f''(x)$.

c) Where is f concave up? Where is concave down? What are the inflection points of f ?

d) Sketch a possible graph of f .

6. Suppose f is an unknown function that's continuous on $(-\infty, \infty)$ and $f'(x) = x^3 - 8$.
- a) Where is f increasing? Where is f decreasing? What are the turning points of f ?

It turns out that $f''(x) = 3x^2$. (For extra practice, use the definition of the derivative to show this!)

- b) Where is f concave up? Where is concave down? What are the inflection points of f ?

- c) Sketch a possible graph of f .

2.5 EXAM 1 REVIEW DAY

For in-class activities on Wednesday, October 8.

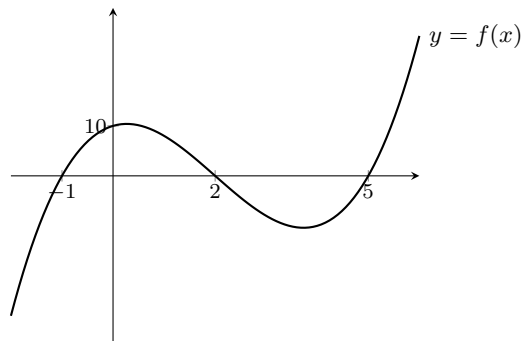
1. a) The picture in (b) shows the graph of a function f . One of these is the value of $f'(-1)$; which one? Explain how you can eliminate the other choices.

-18 -3 3 18

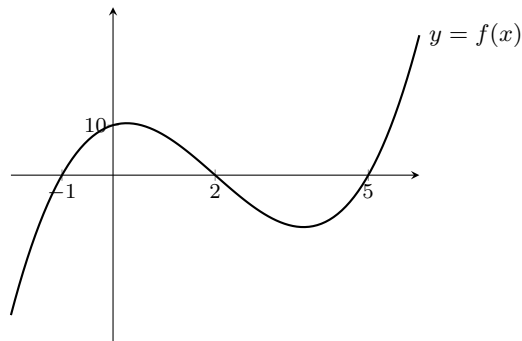
This table shows some additional values of f' :

x	0	1	2	3	4
$f'(x)$	3	-6	-9	-6	3

- b) On the axes below, sketch the graph of $g(x) = 2f(x)$, and label any intercepts.

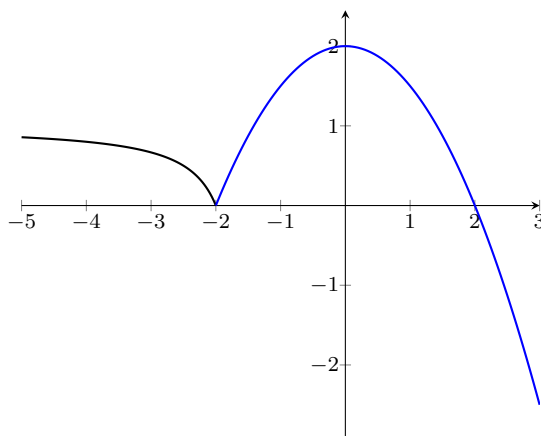


- c) What's $g'(-1)$? Explain.
- d) Here's the graph of f again. On the same axes, sketch the graph of $h(x) = f(2x)$, and label any intercepts.



- e) What's $h'(1)$? Explain.

2. Here's the graph of a function f on $(-5, 3)$:



a) Sketch a graph of f' .

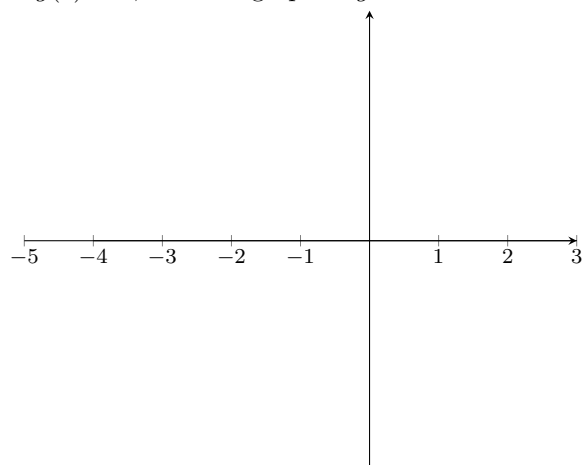
Now, suppose g is a function such that $g'(x) = f(x)$.

b) Where on $(-5, 3)$ is g increasing? Where on $(-5, 3)$ is g decreasing?

c) Where on $(-5, 3)$ is g concave up? Where on $(-5, 3)$ is g concave down?

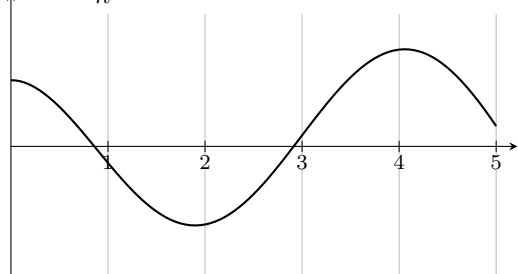
d) What's $g'(-2)$?

- e) If $g(0) = 1$, sketch a graph of g on the axes below.

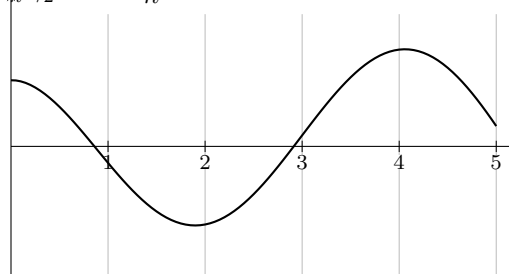


3. In each part, you're given the graph of a function f and an expression. On the graph, sketch a line whose slope is the given expression.

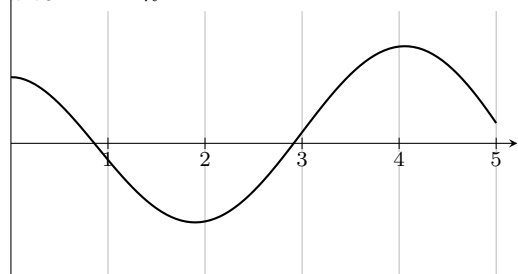
a) $\frac{f(2+h) - f(2)}{h}$ where $h = -0.5$



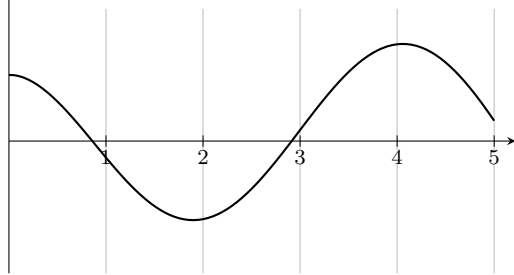
b) $\lim_{h \rightarrow 2} \frac{f(2+h) - f(2)}{h}$



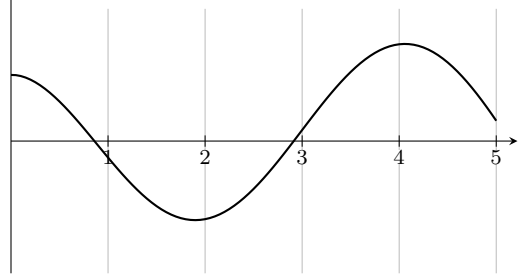
c) $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$



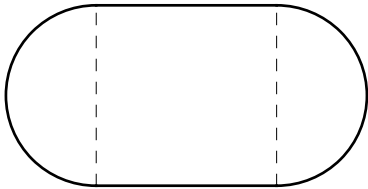
d) $\frac{f(t) - f(3)}{t - 3}$ where $t = 1$



e) $\lim_{t \rightarrow 3} \frac{f(t) - f(3)}{t - 3}$



4. Harvard is building a new running track which is to be 400 meters long. It is to be the perimeter of a region obtained by putting two semicircles on opposite ends of a rectangle, as shown in the picture below (the track is the heavy dark line).



- a) Express the area enclosed by the track as a function of the radius of the ends.

- b) What is the domain of the function in (a)?

- c) The curved portions of the track will cost \$5 per meter to build, and the straight portions will cost \$3 per meter. Express the cost of the track as a function of the radius of the ends.

- d) What is the domain of the function in (c)?

5. Let $f(x) = 3 + \frac{1}{x-2}$.

- a) Use the definition of the derivative to calculate $f'(1)$.

b) Sketch the graph of f . Label all asymptotes, and find and label all intercepts.

c) Are your answers to (a) and (b) consistent with each other?

d) Put the following quantities in increasing order. Justify your ordering.

$$f'(3) \quad f'(4) \quad f'(5)$$

6. Find the following limits algebraically. If the limit does not exist, can you say anything about the one-sided limits?

a) $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x + 3}$

b) $\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3}$

c) $\lim_{x \rightarrow -3} \frac{x + 1}{x + 3}$

UNIT 3

LIMITS AND CONTINUITY

3.1 LIMITS

You should be able to:

- Calculate two-sided and one-sided limits of functions given algebraically or graphically, and understand why such a limit might not exist.
- Recognize when we need to factor and cancel to calculate a limit (for example, with $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$).
- Explain precisely what we mean when we write $\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$. In particular, you should understand that, in both of these cases, $\lim_{x \rightarrow a} f(x)$ does not exist, but that saying $\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$ gives more information than just saying “ $\lim_{x \rightarrow a} f(x)$ does not exist” does.
- Use algebraic methods to determine whether a limit is $\pm\infty$.

1. **Warm up.** In our quest to calculate the instantaneous rate of change, we discovered that it’s useful to reason about what value the average rate of change is approaching as the endpoints of a given input interval approach each other. We’ll review that here before taking a look at limits more generally.

a) If g is a function, what’s the definition of the derivative $g'(7)$?

b) If $g(x) = x^2$, calculate $g'(7)$.

Limits. You can think of the notation $\lim_{w \rightarrow a} f(w)$ as shorthand for:

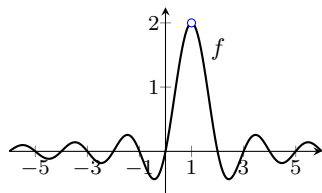
“What number (if any) does $f(w)$ approach as w gets really close to a , without actually being equal to a ?”

This number is called the limit of $f(w)$ as w approaches a . When $f(w)$ approaches a finite value as w approaches a , we say that the limit exists and is finite. This is a pretty general statement.

More precisely, saying $\lim_{w \rightarrow a} f(w) = L$ means that we can make the values of $f(w)$ as close as we like to L by making w sufficiently close to (but not equal to) a .

We'll want to be comfortable reasoning about limits graphically and algebraically. In the following problem, you'll practice finding the value of a limit when our information about the function is given to us graphically.

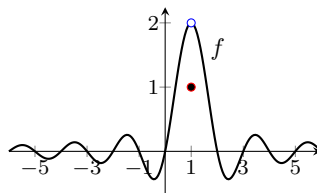
2. a)



$$\lim_{t \rightarrow 1} f(t) = \underline{\hspace{2cm}}$$

$$f(1) = \underline{\hspace{2cm}}$$

b)



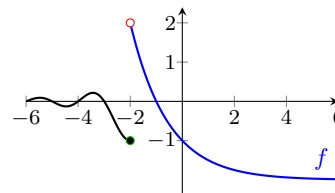
$$\lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm}}$$

$$f(1) = \underline{\hspace{2cm}}$$

$$\lim_{x \rightarrow -3} f(x) = \underline{\hspace{2cm}}$$

$$f(-3) = \underline{\hspace{2cm}}$$

c)



$$\lim_{q \rightarrow -2} f(q) = \underline{\hspace{2cm}}$$

$$f(-2) = \underline{\hspace{2cm}}$$

3. One of the limits above does not exist. Identify which limit this is and summarize what is occurring that makes it so that the limit does not exist.

4. Now we'll switch to evaluating limits when we have the algebraic representation (formula) for a function. A first step is to determine the structure of the limit. Is it a fraction with a numerator and denominator? A difference? A product? See if you can determine its value by reasoning about the parts' individual behavior. Sometimes that is conclusive enough to determine the value of the limit right away. When that does not immediately work, we can sometimes use algebra to simplify the limit into a form where we are then able to reason about its value. Try this approach on the following limits.

a) $\lim_{w \rightarrow 2} \frac{3w + 1}{w}$

b) $\lim_{h \rightarrow 0} \frac{h^2 + 14h}{h}$

c) $\lim_{q \rightarrow 2} \frac{2q - 4}{q^2 - 4}$

Reasoning about limits: The process of determining the structure of a limit and reasoning about the behavior of its parts is really useful because this information can be used to determine the overall limit. Consider the situation where for two real numbers L and M , $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Under these conditions, the following hold:

$$\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L \pm M$$

$$\lim_{x \rightarrow a} (c \cdot f(x)) = c \cdot \lim_{x \rightarrow a} f(x) = c \cdot L, \text{ for any constant } c$$

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M$$

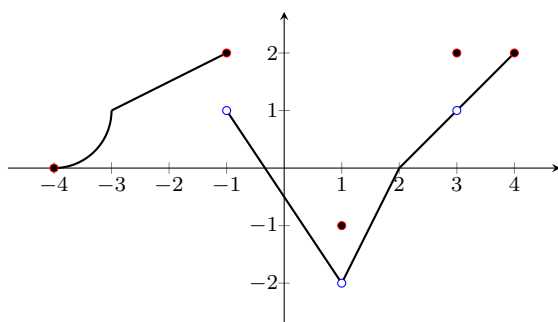
$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}, \text{ provided } M \neq 0$$

Our work earlier demonstrates why we need to specify the direction of approach in some situations.

If a is a number, w can get really close to a from two sides, the left and the right. So, we also look at one-sided limits:

- The limit of $f(w)$ as w approaches a from the left, $\lim_{w \rightarrow a^-} f(w)$, is the number that $f(w)$ approaches as w gets really close to a , while remaining slightly *less* than a .
- The limit of $f(w)$ as w approaches a from the right, $\lim_{w \rightarrow a^+} f(w)$, is the number that $f(w)$ approaches as w gets really close to a , while remaining slightly *greater* than a .
- Notice that the value of a limit is not necessarily equal to the function value at the relevant input value.

5. **Looking at one- and two-sided limits graphically.** The graph of a function $g(x)$ is given below.



Evaluate the following, or explain why they do not exist.

- | | | |
|------------------------------------|-------------------------------------|------------------------------------|
| a) $\lim_{x \rightarrow 1^+} g(x)$ | e) $\lim_{x \rightarrow -4^+} g(x)$ | i) $\lim_{x \rightarrow 3} g(x)$ |
| b) $\lim_{x \rightarrow 1^-} g(x)$ | f) $\lim_{x \rightarrow -3} g(x)$ | j) $g(3)$ |
| c) $\lim_{x \rightarrow 1} g(x)$ | g) $\lim_{x \rightarrow -1} g(x)$ | k) $\lim_{x \rightarrow 4^-} g(x)$ |
| d) $g(1)$ | h) $g(-1)$ | |
6. a) Sketch an example of a function f for which $\lim_{x \rightarrow 2^-} f(x) = 1$ and $\lim_{x \rightarrow 2^+} f(x) = 3$. Can you sketch an example which also has the property that $\lim_{x \rightarrow 2} f(x)$ exists?
- b) **Important Conclusion.** Generalize: In order for $\lim_{x \rightarrow a} f(x)$ to exist, what needs to happen with the one-sided limits $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$?

7. Sketch the graph of $f(x) = \frac{1}{x}$, and use it to find the following limits.

a) $\lim_{x \rightarrow 0^+} \frac{1}{x}$

b) $\lim_{x \rightarrow 0^-} \frac{1}{x}$

c) $\lim_{x \rightarrow 0} \frac{1}{x}$

How could you have found these limits algebraically?

8. Find the following limits algebraically.

a) $\lim_{x \rightarrow 2^-} \frac{x-5}{2-x}$

b) $\lim_{x \rightarrow 2^+} \frac{x-5}{2-x}$

c) $\lim_{x \rightarrow 2} \frac{x-5}{2-x}$

A Note on Notation: In the situation where the limit of a function does not approach a finite value, we ask that you be as specific as possible about what is happening instead:

- In the instance where the function is increasing without bound, we write ∞
- If the function is decreasing without bound, we write $-\infty$
- If neither of the above scenarios is occurring and the limit does not exist, we write "DNE"

9. **Practice Problems.** Find the following limits algebraically. If the limit does not exist, can you say anything about the one-sided limits?

a) $\lim_{t \rightarrow 3} \frac{t-4}{t-3}$

b) $\lim_{t \rightarrow 4} \frac{t-4}{t-3}$

c) $\lim_{x \rightarrow 1} \frac{x^2 - 6x + 8}{x - 2}$

d) $\lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x - 2}$

10. Define $f(x) = \begin{cases} x^2 & \text{if } x \leq 3.9999 \\ \frac{1}{2}x + 3 & \text{if } 3.9999 < x < 4.0001 \\ \frac{1}{x} & \text{if } x \geq 4.0001 \end{cases}$

What is $\lim_{x \rightarrow 4} f(x)$?

3.2 END BEHAVIOR

You should be able to:

- Evaluate $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ and interpret such limits graphically.
- Find the horizontal asymptotes of a function $f(x)$ if given an algebraic formula for $f(x)$.

Warm up. In order for $\lim_{x \rightarrow a} f(x)$ to exist, the following must be true:

Last class, we focused on using limits to describe function behavior as the input variable nears a finite value. Today, we're going to shift our focus to **using limits to describe what we call "end behavior,"** or what happens to the function as the input variable grows or decreases without bound. We notate this by using $x \rightarrow \pm\infty$ so we are evaluating limits, such as $\lim_{x \rightarrow \pm\infty} f(x)$. In conversation, these are sometimes referred to as "limits at infinity."

1. Sketch the graph of $f(x) = \frac{1}{x}$. Based on the graph, what do you think the following limits are?

- a) $\lim_{x \rightarrow \infty} \frac{1}{x}$
- b) $\lim_{x \rightarrow -\infty} \frac{1}{x}$

If you didn't have the graph, how could you have found these limits algebraically?

2. Calculate the following limits.

- a) $\lim_{x \rightarrow \infty} 3x^2 - 10^{100}x - 7$

What information did you use to reason about this limit?

When reasoning about limits to infinity, we want to make use of two big ideas:

- that one term "eventually grows much faster than" another, which is represented by the symbol " $\gg$ "
- that one function "behaves like" another function, which is represented by the symbol " $\sim$ ".

Complete reasoning, (which you'll be asked to show in the future) requires: (1) identifying the fastest growing term in different parts of the function, (2) identifying which simplified function the given function behaves like, (3) computing the simplified function's limit, which computes our original function's limit.

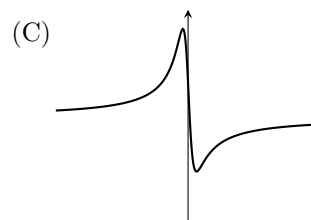
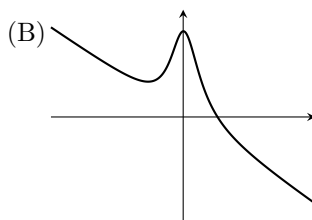
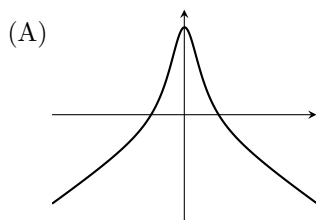
b) $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{4x^2 + 1}$

c) $\lim_{x \rightarrow -\infty} \frac{3x^2 - 2x + 1}{4x^2 + 1}$

d) $\lim_{x \rightarrow \infty} \frac{x + 50 - 3x^3}{4x^2 + 7}$

e) $\lim_{x \rightarrow -\infty} \frac{x + 50 - 3x^3}{4x^2 + 7}$

Which picture below shows the graph of $f(x) = \frac{3x^2 - 2x + 1}{4x^2 + 1}$? How about $g(x) = \frac{x + 50 - 3x^3}{4x^2 + 7}$?



Horizontal asymptotes. Because horizontal asymptotes are a possible feature of a function's end behavior, we can determine whether a function has horizontal asymptotes by evaluating its limits as x increases or decreases without bound. **In calculus-speak, we determine whether a function $f(x)$ has horizontal asymptotes by evaluating $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.** Visually, horizontal asymptotes give the graph of the function the appearance of pulling closer to a horizontal line as we move to the right or left.

3. Does $f(x) = \frac{3x^2}{4x^3 + 1}$ have any horizontal asymptotes? If so, find all.

4. a) Find $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{4x^2 + 1}}$.

b) Is $f(x) = \frac{x}{\sqrt{4x^2 + 1}}$ even, odd, or neither?

c) Based on your answers to (a) and (b), how many horizontal asymptotes does $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{4x^2 + 1}}$ have?

5. Mixed practice with different limits.

a) Evaluate the following limits. If the limit does not exist, can you say anything about the 1-sided limits?

i. $\lim_{x \rightarrow -\infty} \frac{2x - 6}{x^2 - 3x}$

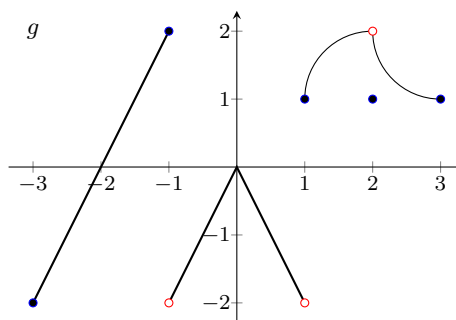
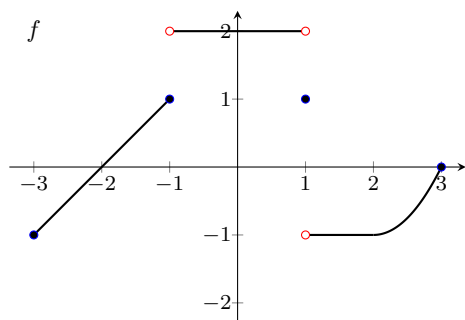
ii. $\lim_{x \rightarrow 1} \frac{2x - 6}{x^2 - 3x}$

iii. $\lim_{x \rightarrow 0} \frac{2x - 6}{x^2 - 3x}$

iv. $\lim_{x \rightarrow 3} \frac{2x - 6}{x^2 - 3x}$

b) Sketch the graph of $f(x) = \frac{2x - 6}{x^2 - 3x}$, and use it to check your answers.

6. The graphs of f and g are given below.



Evaluate each of the following quantities. Even though information about these functions is communicated through graphs, your process for evaluating limits should stay the same.

a) $\lim_{x \rightarrow 2} g(x)$

f) $\lim_{x \rightarrow 1^+} \frac{f(x)}{g(x)}$

b) $g(2)$

g) $\lim_{x \rightarrow -2} \frac{g(x) - g(-2)}{x - (-2)}$

c) $\lim_{x \rightarrow 2} \frac{f(x)}{g(x)}$

h) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$

d) $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$

i) $\lim_{x \rightarrow -1} \frac{f(x)}{g(x)}$

e) $\lim_{x \rightarrow 1^-} \frac{f(x)}{g(x)}$

j) $\lim_{x \rightarrow 1} [f(x) + g(x)]$

7. What are important things to watch out for when calculating limits? Useful strategies?

8. a) Find the following limits.

i. $\lim_{x \rightarrow 2^+} \left(3 - \frac{4}{x-2} \right)$

ii. $\lim_{x \rightarrow 2^-} \left(3 - \frac{4}{x-2} \right)$

iii. $\lim_{x \rightarrow \infty} \left(3 - \frac{4}{x-2} \right)$

b) Sketch the graph of $3 - \frac{4}{x-2}$. Are your answers above consistent with the graph?

3.3 CONTINUITY & REASONING ABOUT SIGNS

You should be able to:

- Explain the idea of continuity, both intuitively and precisely using the limit definition.
- Use limits to classify discontinuities as *removable discontinuities*, *jump discontinuities*, or *vertical asymptotes*, and describe what each type of discontinuity looks like graphically.
- Explain what relationships there are between whether a function is continuous at a and whether it's differentiable at a .
- Take discontinuities into account when constructing a sign chart for a function.

Over the course of the first few problems, we'll be filling in information in the following box. It's full of important definitions and properties!

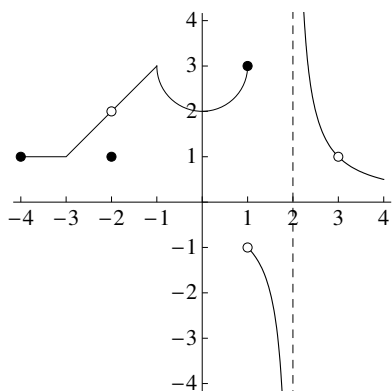
Limit definitions of continuity / discontinuities.

1. A function $f(x)$ is continuous at $x = a$ if _____.
2. What are the 3 types of discontinuities we named, and how do we describe each in terms of limits?
 -
 -
 -

1. a) Using white chalk, sketch some functions $f(x)$ for which $\lim_{x \rightarrow 3} f(x) \neq f(3)$. Try to draw a few examples that look really different!

- b) Using colored chalk, sketch some functions $f(x)$ for which $\lim_{x \rightarrow 3} f(x) = f(3)$. Try to draw a few examples that look really different!

2. Here's the graph of a function $f(x)$.



a) At what values of x in $(-4, 4)$ is $f(x)$ *not* continuous? In other words, at what values of x does $f(x)$ have a discontinuity? What type is each discontinuity?

b) At what values of x in $(-4, 4)$ is $f(x)$ *not* differentiable?

A useful fact. Any function we can write down with a single (non-piecewise) algebraic formula is continuous at every point in its domain. In other words, the discontinuities of a function described by a single formula are where it's undefined.

3. Let $f(x) = \frac{x-3}{x-2}$. Find all values of x in $(-\infty, \infty)$ at which f is discontinuous (by determining where it is not defined), and use limits to classify each (that is, to decide whether it's removable, a jump, or a vertical asymptote).

4. **Important implications!** Decide whether each of the following statements is true or false. If it's false, give an example that shows that it's false (such an example is known as a *counterexample*).

a) If $f(x)$ is continuous at $x = a$, then $f(x)$ is differentiable at $x = a$.

b) If $f(x)$ is differentiable at $x = a$, then $f(x)$ is continuous at $x = a$.

5. a) Can you sketch a function $g(x)$ defined on $[-4, 3]$ which satisfies all 4 of the following properties, or is it impossible?

- $g(-4) = 2$
- $g(3) = -1$
- g has no zeros in $[-4, 3]$
- g is continuous on $[-4, 3]$

b) Can you sketch a function $g(x)$ defined on $[-4, 3]$ which satisfies the first 3 properties in (a), or is it impossible?

6. **Summarize.** When we are working with continuous functions, which happens throughout calculus, some special properties hold, which we will frequently take advantage of.

a) If a function g switches from positive to negative or vice versa at $x = a$, what can you say about the behavior of g at $x = a$?

This is a specific application of a broader theorem, known as the **Intermediate Value Theorem**, which states that if some function f is continuous on the closed interval $[a, b]$ and $f(a) = A$ and $f(b) = B$, then somewhere on the interval, f attains every value between A and B . (Gottlieb, p.271)

b) **Key Skill:** There are many situations where we want to determine the sign of a function. If you want to make a **sign chart** for $g(x)$ and $g(x)$ is given by a single (non-piecewise) algebraic formula, you should start by finding values of x where $g(x)$ is _____.

7. **Practice using sign charts.** Suppose f is a function with domain $(-\infty, \infty)$, and $f'(x) = \frac{x+2}{\sqrt[3]{x}}$. Where is f decreasing? Build a sign chart to determine the answer.

8. **Practice identifying and classifying discontinuities.** Let $f(x) = \frac{x^2 + 2x}{|x|}$. Find all values of x at which $f(x)$ is discontinuous. For each such value a , calculate $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$, and use this to classify the discontinuity a .

9. Let $f(x) = \begin{cases} x^{2/3} & \text{if } x < 8 \\ x - 4 & \text{if } 8 \leq x \leq 16 \\ \frac{1}{x - 20} & \text{if } x > 16 \end{cases}$

Since f isn't defined by a single formula, what points should we consider if we want to find the discontinuities of f ? For each such value a , use limits to determine whether f is continuous at a . If f is not continuous at a , determine which type of discontinuity f has there.

UNIT 4

DERIVATIVE RULES, APPROXIMATION, AND EXTREMA

4.1 THREE IMPORTANT DIFFERENTIATION RULES

You should be able to:

- Explain why the Power Rule is true for positive integer exponents.
- Explain why the Constant Multiple Rule and Sum Rule are true.
- Use the Power Rule, Constant Multiple Rule, and Sum Rule, and recognize when they do and do not apply.

Civilization advances by extending the number of important operations which we can perform without thinking about them.

– Alfred Whitehead, mathematician and philosopher

We've previously found the derivatives of several functions; fill in the empty boxes. (You should be able to do these by thinking graphically.) Also, rewrite the last two derivatives in the form ax^b .

Function	7	x	$5x + 3$	x^2	$x^{-1} = \frac{1}{x}$	$x^{-2} = \frac{1}{x^2}$
Derivative				$2x$	$-\frac{1}{x^2} = \underline{\hspace{2cm}}$	$-\frac{2}{x^3} = \underline{\hspace{2cm}}$

Many of these results came from calculating the limit definition of the derivative, a labor intensive endeavor, for sure! In today's class we'll start by coming up with a general rule for the derivative of this family of functions.

1. We'd like to find the derivative of x^n where n is a positive integer. Write out the limit definition of $\frac{d}{dx}(x^n)$. (In other words, if $f(x) = x^n$, write out the limit definition of $f'(x)$.)

2. **Exploring** $(x + h)^n$.

We know $(x + h)^0 = \underline{\hspace{1cm}}$ and $(x + h)^1 = \underline{\hspace{1cm}}$. Now, expand:

a) $(x + h)^2$

b) $(x + h)^3$ (Can you make use of what you just found out about $(x + h)^2$?)

- c) Continuing on in this way seems tiring, so let's be more strategic. First, let's organize what we've figured out so far. Fill in the blanks with the coefficients you've found (we don't know the last row yet):

$$\begin{aligned}
 (x+h)^0 &= \mathbf{1} \\
 (x+h)^1 &= \mathbf{1x + 1h} \\
 (x+h)^2 &= \underline{\hspace{1cm}}x^2 + \underline{\hspace{1cm}}xh + \underline{\hspace{1cm}}h^2 \\
 (x+h)^3 &= \underline{\hspace{1cm}}x^3 + \underline{\hspace{1cm}}x^2h + \underline{\hspace{1cm}}xh^2 + \underline{\hspace{1cm}}h^3 \\
 (x+h)^4 &= \underline{\hspace{1cm}}x^4 + \underline{\hspace{1cm}}x^3h + \underline{\hspace{1cm}}x^2h^2 + \underline{\hspace{1cm}}xh^3 + \underline{\hspace{1cm}}h^4
 \end{aligned}$$

Find the coefficient of x^2h^2 in $(x+h)^4$. Which coefficients from $(x+h)^3$ were relevant?

Find the coefficient of x^3h in $(x+h)^4$.

Fill in the last row above.

- d) Even writing the table above is kind of tiring, so we often just write the coefficients. In that case, it's called *Pascal's triangle*. Fill in the next few rows of Pascal's triangle.

$$\begin{array}{ccccccc}
 & & & & 1 & & & \\
 & & & 1 & & 1 & & \\
 & & 1 & & 2 & & 1 & \\
 & 1 & & 3 & & 3 & & 1 \\
 & _ & _ & _ & _ & _ & _ & \\
 & _ & _ & _ & _ & _ & _ & _ \\
 & _ & _ & _ & _ & _ & _ & _ \\
 & _ & _ & _ & _ & _ & _ & _
 \end{array}$$

What's $(x+h)^7$?

- e) *Just for fun!* Instead of writing down the coefficients of Pascal's triangle, do the following: if the coefficient is odd, color the corresponding hexagon black; otherwise, leave it blank.

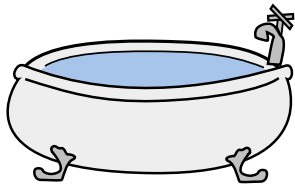


3. Use the definition of the derivative to find $\frac{d}{dx}(x^4)$. (In other words, if $f(x) = x^4$, find $f'(x)$.)
4. If n is any positive integer, use the limit definition of the derivative to find $\frac{d}{dx}(x^n)$.
5. Do the derivatives of x^{-1} and x^{-2} fit the rule you came up with in #4? How about the derivative of x^0 ?

Power Rule: $\frac{d}{dx}(x^n) =$

The Power Rule is true for all n , including negative values and non-integer values like -0.5 and π , but we've only explained why for positive integers n . Nonetheless, you may use it for all n .

6.



At noon one day, Becky and Gregg start making a special blend of purple Gatorade in Becky's bathtub. Let $f(t)$ be the volume (in liters) of blue Gatorade Becky has poured into the tub t minutes after noon, and let $g(t)$ be the volume in liters of red Gatorade Gregg has poured in t minutes after noon.

a) What does $f + g$ represent?

b) What does $(f + g)'$ represent? What are its units? How does it relate to f' and g' ?

c) Let $h(t)$ be the amount in gallons of blue Gatorade that Becky has poured in t minutes after noon. Write an expression for $h(t)$ in terms of $f(t)$. (There are 0.26 gallons in a liter.)

d) What are the units of $h'(t)$? How does $h'(t)$ relate to $f'(t)$?

Sum Rule: $\frac{d}{dx}[f(x) + g(x)] =$

Constant Multiple Rule: If c is a constant, then $\frac{d}{dx}[cf(x)] =$

7. **Practice!** Use our new rules to find the derivatives of the following functions.

a) $g(x) = x^5 + 5x^4 + 7$

b) $g(x) = \frac{5}{\sqrt[3]{x}} - \frac{1}{4x} + \sqrt{x^5} + \pi^6$

c) $h(x) = \frac{x^2 - 1}{\sqrt{x}}$

8. What's the slope of the line tangent to $y = \frac{x^2 - 1}{\sqrt{x}}$ (the function in #7(c)) at $x = 4$? How would we express this quantity using function notation?
9. Let $g(x) = x^5 + 5x^4 + 7$ (the same function as in #7(a)).
- On what intervals is $g(x)$ increasing? On what intervals is it decreasing? What are its turning points?
 - On what intervals is $g(x)$ concave up? On what intervals is it concave down? What are its inflection points?
 - What is the y -intercept of the graph of g ? Is it easy to find the x -intercepts?
 - What's $\lim_{x \rightarrow \infty} g(x)$? $\lim_{x \rightarrow -\infty} g(x)$?
 - Sketch a rough graph of $g(x)$ on $[-6, 6]$. (Note: $g(-6) = -1289$ and $g(6) = 14263$.)

10. **More practice!** Evaluate the derivative of each of the following functions, and simplify your answers. There's one function that you won't be able to differentiate using the rules we've learned so far; which one is it?

a) $f(x) = (x^5 - 3x^2 - 4)\sqrt[3]{x^2}$

b) $r(x) = \sqrt{x^5 - 3x^2 - 4}$

c) Given $f(t) = (t^2 + 1)^2$, find $f'(1)$

4.2 THE PRODUCT & QUOTIENT RULES, TANGENT LINE APPROXIMATION

You should be able to:

- Explain why the Product and Quotient Rule are true, and use them (together with the rules we've learned before) to differentiate functions.
- Identify functions that can and can't be differentiated using just the rules we've learned so far.
- Use secant lines and tangent lines to approximate values like $\sqrt{22}$, and use concavity to determine whether your approximations are upper or lower bounds.

Warm-up/Discussion. Looking back at your last assignment offers us the chance to explore a critical use of the derivative in approximating function values that can't be computed exactly. We'll be taking a look at this in the second half of our lesson, and keeping this in mind will be really helpful.

1. At noon one day, Jack plants a magical bean. To his delight, it grows quickly into a giant beanstalk. Let $f(t)$ be the beanstalk's height in feet t hours after noon.

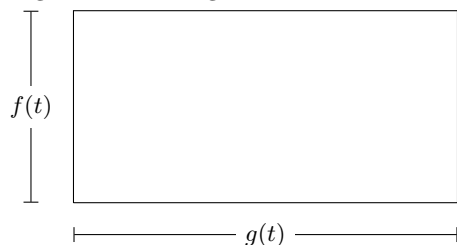
If h is a fairly small positive number, roughly how much does the beanstalk grow between t hours after noon and $t + h$ hours after noon?

2. Suppose we have two differentiable functions f and g , and let $P(t) = f(t)g(t)$ be their product. We'll be exploring what the derivative of this product is in the following parts.

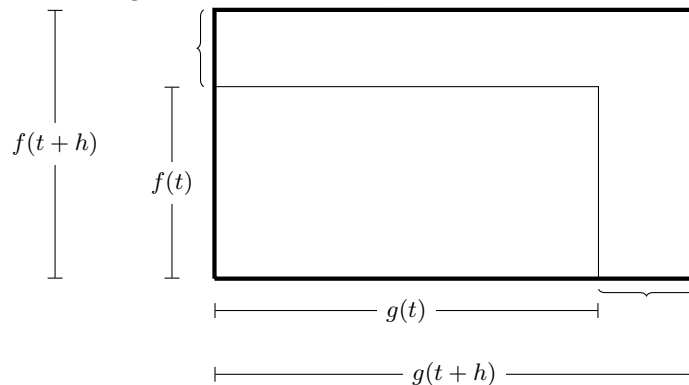
a) According to the limit definition of the derivative, what is $P'(t)$?

b) Can you think of a picture to draw to represent a product?

- c) Imagine a rectangle that's growing over time. Let $f(t)$ and $g(t)$ be its height and width at time t . Imagine looking at the rectangle at time t :



and then again a short time later, at time $t + h$:



- i. Approximate the lengths marked by the braces. (Hint: How is #1 relevant?)
- ii. How would you picture $P(t+h)$ in this picture? How about $P(t)$? Use this to find $P'(t)$.

Product Rule: $\frac{d}{dx}[f(x) \cdot g(x)] =$

3. Evaluate $\frac{d}{dx} [(x^4 + x^7)(3x^7 + x^2)]$.

Quotient Rule (*you'll show this on homework!*): $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}$

A way to remember it: <https://youtu.be/jwuiVb84Xx4>

4. If $u(r) = \frac{5r + 3}{2r^7 + 1}$, find $\frac{du}{dr}$.

5. **Practice!** Find the derivative of each function without using the Quotient Rule. (There are lots of notations for finding derivatives; this problem shows you several.)

a) If $y = \left(\frac{1}{x} - 7x + 4 \right) (x^3 - x + 3)$, find y' .

b) If $y = \frac{7t^2 + 3t + 15}{t\sqrt{t}}$, find $\frac{dy}{dt}$.

c) If $h(w) = \frac{5w^2 + 4w + 3}{\pi}$, find $h'(w)$.

6. Which of the following can you differentiate using the rules we've learned? Which rules would you use for the ones you can differentiate?

a) $a(s) = (3s + 1)^2$

b) $b(s) = (3s + 1)^{1/2}$

c) $c(s) = (3s + 1)^{-2}$

7. Here we'll explore using the derivative to execute what we call a **tangent line approximation**.
- Use an appropriate tangent line to approximate $\sqrt{22}$. Is your estimate an upper bound or lower bound for the actual value of $\sqrt{22}$?
 - What's another way you could approximate $\sqrt{22}$? Would that give you an upper bound or lower bound for the actual value of $\sqrt{22}$?
 - What was the relevant function $f(x)$ in this problem? What does the sign of $f''(x)$ have to do with your answer?

8. In your Edfinity homework, you saw that the derivative of $f(x)^2$ is not $2f(x)$. Now, we'll find a correct formula for the derivative of $f(x)^2$.

Let $g(x) = f(x)^2$. Find $g'(x)$. (Which rule do you need to use?)

9. We'd like to get an accurate graph of $f(x) = \frac{18x^2 + 2}{x}$. Answer the questions below in whatever order makes most sense to you, and use them to sketch the graph of f .

a) What is the domain of f ? Does f have any useful symmetry?

b) Can you find the x - and y -intercepts of f ?

c) Find and classify all discontinuities of f . Justify your classifications using limits. If a limit does not exist, find both one-sided limits.

d) Find all horizontal asymptotes of f .

- e) On what intervals is $f(x)$ increasing? On what intervals is it decreasing?
- f) On what intervals is $f(x)$ concave up? On what intervals is it concave down?
- g) What are the turning points of f ? The inflection points of f ?
- h) Sketch a rough graph of $f(x)$ that incorporates all of the above information.

4.3 THE CHAIN RULE

You should be able to:

- You should be able to use the Chain Rule to find the derivative of the composition of two functions.
- You should understand why the Chain Rule makes sense in a real-world problem (such as #1 on the worksheet “The Chain Rule”).
- You should be comfortable looking at a function and determining its structure: is it a product? A composition? (If so, what is the outer function?)

1. **Warm-up:** On your problem set last night, you reviewed compositions of functions. Decompose each of the functions below so that $h(x) = f(g(x))$.

a) $h(x) = (10x^2 + 4)^2$ can be written as $f(g(x))$ with $g(x) = 10x^2 + 4$ and $f(u) = \underline{\hspace{2cm}}$

b) $h(x) = (2x + 5)^{1/2}$ can be written as $f(g(x))$ with $g(x) = \underline{\hspace{2cm}}$ and $f(u) = \underline{\hspace{2cm}}$

c) $h(x) = \frac{1}{3x}$ can be written as $f(g(x))$ with $g(x) = 3x$ and $f(u) = \underline{\hspace{2cm}}$

d) $j(x) = \frac{1}{(2x + 5)^{1/2}}$ can be written as $f(g(h(x)))$

with $h(x) = \underline{\hspace{2cm}}$, $g(u) = \underline{\hspace{2cm}}$ and $f(v) = \underline{\hspace{2cm}}$

2. Composition of functions allows us to make new models from old ones without starting from scratch. Here is an example: A rainwater collection tank is shaped so that when the tank is filled with rainwater, the height of water in the tank can be measured.

a) Which quantities are varying in this problem?

b) Right away this description relates which two quantities and how? (Fill in the blank.)

$\underline{\hspace{2cm}}$ depends on $\underline{\hspace{2cm}}$.

c) Consider what happens in a rainstorm. As time passes, describe how we would expect the quantities to change?

(Fill in the blank.) This now introduces time. How are these three quantities related?

$\underline{\hspace{2cm}}$ depends on $\underline{\hspace{2cm}}$, which now depends on $\underline{\hspace{2cm}}$.

d) Let's explore this connection and how it relates these quantities' rates of change: if you know that at a given moment in time, rainwater is pouring into the tank at 2 mL/s and you know that each additional mL causes the height of the water to increase by 0.2 mm, determine how height is changing as time passes.

- e) This connection between the instantaneous rates of change of functions in a composition is known as the **Chain Rule**. There are a few ways we use notation to describe this relationship.

Fill in the blank, then annotate with each symbol's meaning: Given that height h , was a function of volume v , and v was a function of time t , we can calculate the instantaneous rate of change of height h with respect to time t :

$$\frac{dh}{dt} = (\text{_____}) \cdot (\text{_____})$$

Let $h = f(v)$, that is, height is a function of volume. Let $v = g(t)$, that is, volume is a function of time. Given these formal definitions, we could now describe the relationship between height and time as a composition, written $h = f(g(t))$. In which case, the same calculation to find how height is changing with respect to time could be written:

$$\frac{d}{dt}f(g(t)) = \text{_____} \cdot \text{_____}$$

The Chain Rule: Given a composition of functions $f(g(t))$, the derivative is:

$$\frac{d}{dt}[f(g(t))] = \text{_____} \quad \text{or} \quad \frac{df}{dt} = \text{_____}$$

3. This allows us the ability to differentiate more efficiently than before and tackle new functions.

a) From earlier in the week: $a(s) = (3s + 1)^2$. Find $a'(s)$ without first expanding.

b) $k(r) = \frac{1}{2r^7 + 1}$. Find $k'(r)$.

c) $h(x) = \pi^3(7x + 3)^{10}$

d) $y = \frac{\sqrt{x^2 + 1}}{\pi}$. Find $\frac{dy}{dx}$

e) The Chain Rule can also be used in combination with other rules.

i. From earlier in the week, $u(r) = \frac{5r + 3}{2r^7 + 1}$. Find $\frac{du}{dr}$ without using the Quotient Rule.

ii. Find $f'(x)$ if $f(x) = \frac{x^2}{(x - 2)(3x + 5)}$.

4. a) How does the graph of $(x + 5)^2$ relate to the graph of x^2 ? Use the Chain Rule to differentiate $(x + 5)^2$. Is your answer consistent with the graph and what you know about the derivative of x^2 ?

- b) More generally, if c is a constant, how does the graph of $f(x + c)$ relate to the graph of $f(x)$? What does the Chain Rule tell you about the derivative of $f(x + c)$?

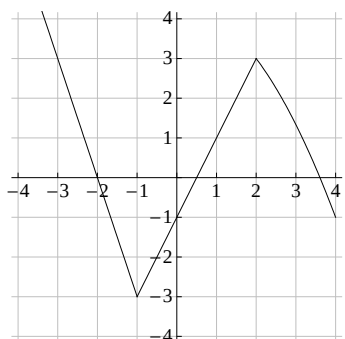
5. You are given the following information about three functions f , g , and h .

$$\begin{array}{lll} h(1) = 2 & g(2) = 3 & f'(3) = 6 \\ h'(1) = 4 & g'(2) = 5 & \end{array}$$

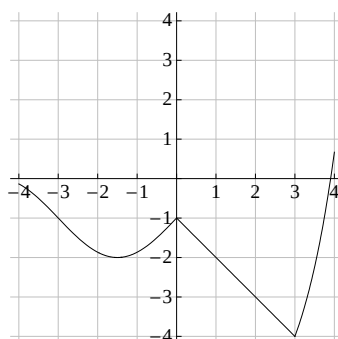
If $r(x) = f(g(h(x)))$, do you have enough information to find $r'(1)$? If so, compute it. If not, what additional information do you need?

6. Here are graphs of two functions, $f(x)$ and $g(x)$. If $h(x) = f(g(x))$, what is $h'(1)$?

$f(x)$



$g(x)$

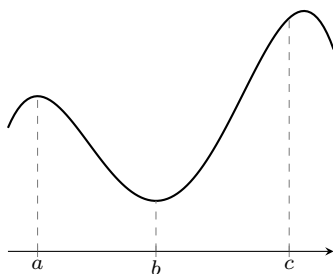


4.4 MAXIMA AND MINIMA OF FUNCTIONS

You should be able to:

- Explain the difference between a *local minimum / maximum* and a *global (or absolute) minimum / maximum*.
- Find the local extrema of a function by first finding *critical points* and then using the First Derivative Test or Second Derivative Test.
- Explain why the First and Second Derivative Tests work.

1. **Warm up.** Here's the graph of a function f .



a) Based on the graph, are each of the following positive, negative, or zero? How can you tell?

i. $f'(a)$

ii. $f'(b)$

iii. $f'(c)$

b) Based on the graph, are each of the following positive or negative? How can you tell?

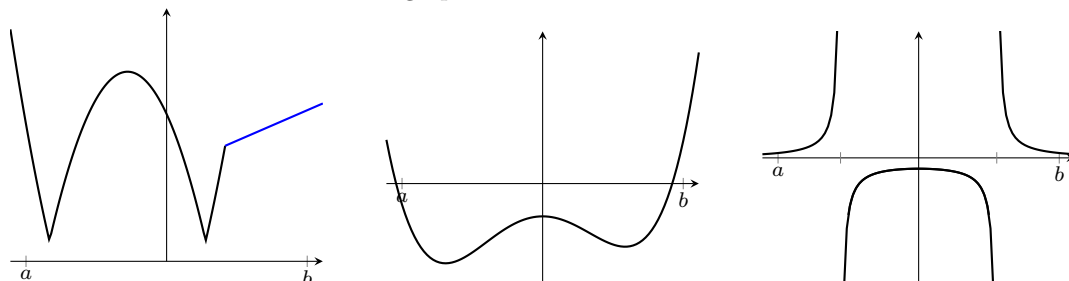
i. $f''(a)$

ii. $f''(b)$

iii. $f''(c)$

These peaks and valleys, known as **extrema** (the broad category that contains maxima and minima) are identifiable based on the behavior of the function's first and second derivatives at a given input. Let's get more technical and explore these definitions.

2. a) Intuitively, a local maximum of f is a point on the graph of f that's at least as high as all nearby points. More precisely, f has a local maximum at c if $f(c) \geq f(x)$ for all x near c . Mark the local maxima and local minima on the graphs below.



- b) We say that a function f has a global maximum (also known as an absolute maximum) at c if $f(c)$ is the highest value of f anywhere in the domain; more precisely, f has a global maximum at c if $f(c) \geq f(x)$ for all x in the domain.

For each graph in (a), does f have a global minimum and a global maximum on $[a, b]$? If so, mark where f achieves its global maximum and global minimum values.

- c) For each of the above graphs, does $f(x)$ have a global minimum and a global maximum on (a, b) ? If so, where?

3. a) Often, our goal is to find the global minimum or maximum (if it exists) of a function on a given interval. A good starting point is to first find the local extrema ("local extremum" is a shorthand way of saying "local minimum or local maximum").

What can you say about $f'(c)$ at a value c where f has a local extremum?

Definition. A critical point of f is a number c in the domain of f such that...

Note: The Gottlieb textbook has a slightly different definition; it also considers endpoints of the domain to be critical points, while we don't.

- b) Is every critical point a local minimum or local maximum?
- c) Besides looking at local extrema, where else should we look to see if a function has global extrema on a given interval?

4. **Finding and classifying local extrema.** In this problem, we'll look at the function $f(x) = x^3 + 3x^2 + 1$.

- a) Find all critical points of f .
- b) Make a sign chart for f' , and use this to decide whether each of the critical points you found is a local minimum, a local maximum, or neither.

What you just did is called the **First Derivative Test** to decide whether a critical point is a local minimum, a local maximum, or neither. Summarize the steps here.

5. **Finding and classifying local extrema.** There's another way to decide whether a critical point is a local min, local max, or neither. That's what we'll look at in this problem!

Let $f(x) = x^4 - 8x^2 + 16$.

a) Find all critical points of f .

b) For each critical point c of f , find the sign of $f''(c)$. What does this tell you about the critical point c ?

c) On what intervals is f increasing? On what intervals is it decreasing? (How little additional work can you do to answer this question?)

6. Let $f(x) = x^3$ and $g(x) = x^4$.

a) Sketch graphs of $f(x)$ and $g(x)$.

b) What is $f'(0)$? $f''(0)$?

c) What is $g'(0)$? $g''(0)$?

Second Derivative Test. Suppose $f'(c) = 0$. What can you conclude if you have the following additional information about $f''(c)$?

- $f''(c) > 0$.
- $f''(c) < 0$.
- $f''(c) = 0$.

7. Let $f(x) = 3x^{2/3} - x$. Find all critical points of f , and classify each as a local minimum, local maximum, or neither. (Do you find the First Derivative Test or Second Derivative Test more useful here?)

8. What are advantages and disadvantages of the Second Derivative Test vs. the First Derivative Test?

9. **Practice interpreting a sign chart to determine global extrema.** Suppose you've made the following sign chart for the derivative of a function $f(x)$. The function $f(x)$ is continuous and differentiable on $(-\infty, \infty)$.

		0	3	5	
sign of f'	+	-	+	-	

- a) Does f have a global maximum on $(-\infty, \infty)$? (Definitely yes, definitely no, or maybe?) If so, where could it be?
- b) Does $f(x)$ have a global minimum on $(-\infty, \infty)$? If so, where could it be?
- c) Does f have a global maximum on $[-2, 10]$? If so, where could it be?
- d) Does f have a global minimum on $[-2, 10]$? If so, where could it be?
10. **Practice determining global extrema.** Let's look again at $f(x) = x^3 + 3x^2 + 1$ from #4.
- a) Does f have a global maximum on $[-4, 0]$? If so, where? How about a global minimum?
- b) Does f have a global maximum on $[-4, 2]$? If so, where? How about a global minimum?
- c) Does f have a global maximum on $(-3, 3)$? If so, where? How about a global minimum?
11. Does $f(x) = x - 2 + \frac{1}{x}$ have a global minimum on $(0, \infty)$? If so, where? Does f have a global maximum on $(0, \infty)$? If so, where?

4.5 FINDING GLOBAL/ABSOLUTE EXTREMA

You should be able to:

- Explain what the Extreme Value Theorem says.
- Determine whether a given function has global extrema on a given domain and, if so, find the global extrema.

1. We'll begin by seeing if we can observe when global extrema are guaranteed to occur. In the following problem, various conditions are described. Can you sketch an example of each of the following, or is it impossible?

a) A function that:

- is continuous on $(-2, 2)$.
- has a global maximum value of 3 and no global minimum on $(-2, 2)$.⁽¹⁾

c) A function that:

- is continuous on $(-2, 2)$.
- has both a global maximum and a global minimum on $(-2, 2)$.

b) A function that:

- is defined on $[-2, 2]$.
- has a global minimum but no global maximum on $[-2, 2]$.

d) A function that:

- is continuous on $[-2, 2]$.
- does not have a global maximum on $[-2, 2]$.

⁽¹⁾The phrase “global maximum value of 3” means the y -value of the global maximum is 3, while the phrase “global maximum at 3” means the x -value is 3.

2. Under what conditions can we be sure that a function f has a global maximum and a global minimum on a given domain?

3. Suppose you've made the following sign chart for the derivative of a function $f(x)$. The function $f(x)$ is continuous and differentiable on $(-\infty, \infty)$.

		0		3		5		
sign of f'		+		-		+		-

- a) Does f have a global maximum on $(-\infty, \infty)$? (Definitely yes, definitely no, or maybe?) If so, where could it be?
- b) Does $f(x)$ have a global minimum on $(-\infty, \infty)$? If so, where could it be?
- c) Does f have a global maximum on $[-2, 10]$? If so, where could it be?
- d) Does f have a global minimum on $[-2, 10]$? If so, where could it be?
- e) Does f have a global minimum on $(-2, 10)$? If so, where could it be?

4. Summarize: What's a strategy for deciding whether a continuous function f has a global min/max on a particular domain? (And, if it has one, how do you find where it is?)

The **Extreme Value Theorem** guarantees the existence of a function's global extrema. Under what conditions does a function $f(x)$ definitely have a global max and a global min?

5. **Apply our strategy to find global extrema.**

- a) Does $f(r) = 2\pi r^2 + \frac{4000\pi}{r}$ have a global maximum and global minimum on $[2, 20]$? (You should be able to answer this question with a yes/no with relatively little work.) If you answer yes, where do these global extrema happen? Give the r -values at which the global minimum and global maximum occur.

- b) Does f have a global minimum and global maximum on $(0, \infty)$? If so, where?

6. *A special case.* Maria is trying to determine whether a continuous function f has a global maximum on $(1, 7]$. She's already determined that the only critical point of f in $(1, 7]$ is 5.

a) If Maria happens to know that 5 is a local minimum, what should she do next to finish deciding whether f has a global maximum on $(1, 7]$?

b) If Maria instead happens to know that 5 is a local maximum, what should she do next to finish deciding whether f has a global maximum on $(1, 7]$?

c) **Summarize:** When can we use a strategy other than that in #4 to find whether / where a continuous function f has a global min / max on a particular domain?

7. A company sells canned tomatoes to restaurants. They want to pack the tomatoes in steel cans. Each can should have a volume of 2000π cubic centimeters. The company's factory can only manufacture cans that are at least 5 cm tall and have a radius of at least 2 cm. In order to conserve resources, the company wants to minimize the amount of steel needed for each can. What dimensions should they make their cans?

8. Let $f(x) = \frac{x}{x^2 + 1}$.

a) Find all critical points of f on $(-\infty, \infty)$. (Note: there are multiple approaches to differentiate this function - which do you prefer?)

b) Classify each critical point as a local maximum, local minimum, or neither. Is it easier to do this using the First Derivative Test or the Second Derivative Test?

c) Does $f(x)$ have a global minimum on $(-\infty, \infty)$? If so, where?

d) Does $f(x)$ have a global maximum on $(-\infty, \infty)$? If so, where?

e) Does $f(x)$ have a global maximum on $[-0.5, 2]$? If so, where?

f) Does $f(x)$ have a global minimum on $[-0.5, 2]$? If so, where?

4.6 OPTIMIZATION

You should be able to:

- Set up and solve optimization problems using calculus.
- Justify that your solution is optimal using appropriate techniques from calculus.

1. a) **Warm up.** Last time, we talked about how to determine whether a continuous function f has a global maximum / minimum on a given domain and, if so, where. Summarize the method.

- b) *Important special case!* Suppose we're maximizing or minimizing a continuous function f on an interval, and f has exactly one critical point in the interval. Are there any other methods we could use to show that f has a global maximum or minimum there?

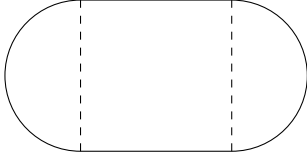
General strategy in optimization problems:

1. Identify the goal. (Like: “Maximize area” or “Minimize cost”)
2. Express the goal quantity as a function of 1 variable.
3. Find critical points.
- 4.
5. Make sure you answer the question!

3. A soda company wants its aluminum soft drink cans to have a volume of 120π cubic centimeters. They need to use stronger aluminum for the tops and bottoms of the cans, and this stronger aluminum costs twice as much per cm^2 as the aluminum used for the sides. If the company wants to minimize the cost of each can, what dimensions should their cans be?

4. Walter is planning to buy a custom-made jewelry box for his mother's next birthday. The bottom of the box will be a square. The sides and bottom will be made out of mahogany, which costs 30 cents per square inch. The top will be made out of maple, which costs 50 cents per square inch. Walter has \$60 to spend on the present and wants to get a box with the largest volume possible. What dimensions should his box be?

5. Cambridge Common is renovating its grounds. They want to include a beautiful fountain surrounded by a walking path. The fountain will be in the shape of a rectangle. There will be a walking path in the park to enjoy the fountain. The path will follow two opposite sides of the fountain and two semi-circles on the other sides of the rectangle. If the path around the fountain is going to be 440 yards long, what dimensions would maximize the area of the fountain?



UNIT 5

EXPONENTIAL FUNCTIONS

5.1 AN INTRODUCTION TO EXPONENTIAL FUNCTIONS

You should be able to:

- Model quantities that have a constant percent change (per unit of input).
- Graph b^x for different values of b (and be able to combine this with what we know about altering functions to graph functions involving b^x).
- Apply exponent rules.

1. **Investment scenario 1.** On his birthday last year, Lin invested \$1,000 in a small company. A year later, he discovers that his investment has grown in value by \$100 so is now worth \$1,100. Suppose that Lin's investment grows at a constant rate.

- a) Fill in the table below.

t	Value of Lin's investment after t years
0	\$1000
1	
2	
3	

- b) Write a function giving the value of Lin's investment after t years. What type of function is this?

- c) By what percent does the value of Lin's investment increase in the first year? The second year? The third?

2. **Investment scenario 2.** Lex also invested \$1,000 last year, but suppose that her investment grows by 10% each year.

a) Is Lin or Lex better off in the long run? How do you know?

b) Fill in the table below. Do you see a pattern? (Hint: If something increases by 10%, it gets multiplied by _____.)

t	Value of Lex's investment after t years
0	\$1000
1	
2	
3	

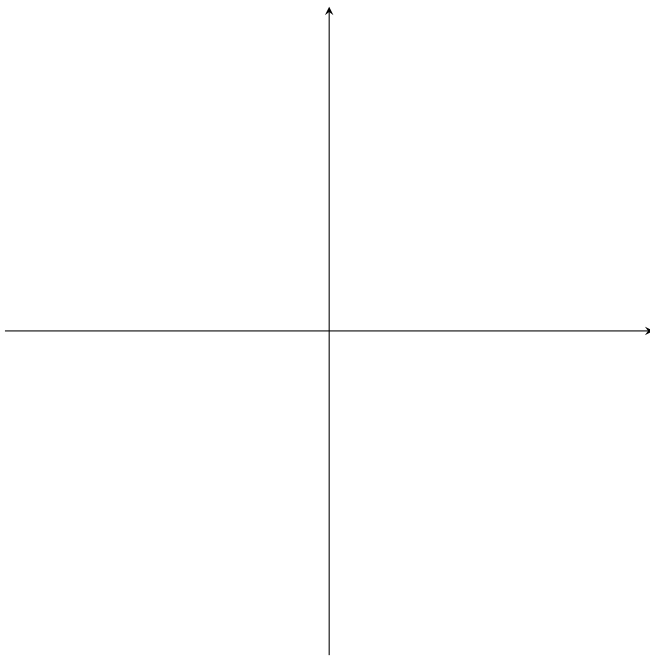
c) Write a function giving the value of Lex's investment after t years.

An exponential function is a function of the form $f(x) = \underline{\hspace{2cm}}$.

- If money is changing linearly over time, that means it changes by the same _____ each year.
- If money is changing exponentially over time, that means it changes by the same _____ each year.

Another way of saying the same thing is that the amount of money is

3. In a certain town in 1975, the amount of whole milk consumed by the average resident was 20 gallons (over the entire year). Since 1975, that consumption has been decreasing by 2% each year.
- (Fill in the blank.) The yearly percent change is (choose one: $\pm$) _____ %.
 - Given this yearly percent change, what percent remains after each year?
 - In order to calculate the percent of milk consumption you found in part (b), what decimal should you multiply by? Why?
 - Write a formula for $M(t)$, the amount of whole milk consumed by the average resident t years after 1975.
4. **The graph of an exponential function.** On the following set of axes, graph $y = 2^x$, $y = 3^x$, and $y = \left(\frac{1}{2}\right)^x$. Label all intercepts and asymptotes. (Also indicate which graph is which.)



5. **The end behavior of an exponential function.** Formalizing the existence of asymptotes, evaluate the following limits:

For values of $b > 1$

$$\lim_{x \rightarrow \infty} b^x = \underline{\hspace{2cm}} \quad \lim_{x \rightarrow -\infty} b^x = \underline{\hspace{2cm}}$$

For values of b between 0 and 1, ($0 < b < 1$)

$$\lim_{x \rightarrow \infty} b^x = \underline{\hspace{2cm}} \quad \lim_{x \rightarrow -\infty} b^x = \underline{\hspace{2cm}}$$

Exponential growth and decay formulas. We call $f(x) = C \cdot b^x$ an exponential growth function if $C > 0$ and b _____, and we call it an exponential decay function if $C > 0$ and b _____.

6. If $f(x) = 2^x$, what can you say about the signs of $f'(x)$ and $f''(x)$?

7. Sketch the graph of $3\left(\frac{4}{3}\right)^x - 5$, and label the y -intercept and any asymptotes. Is this the same as $4^x - 5$?

8. Your friend is having a hard time remembering how exponents work. Help them figure out how to rewrite each of the expressions below, and explain why those rules make sense.

a) $b^x b^y =$

b) $(b^x)^y =$

c) $(ab)^x =$

9. Simplify the following expression as much as possible. (You should be able to write them all so there are no fractions left.)

a) $\frac{(ab^2)^x}{b^x}$

b) $\frac{b^{2x} + b^{3x}}{b^x}$

c) $\frac{C \cdot 4^{t+1} - C \cdot 4^t}{C \cdot 4^t}$

d) $\frac{(3a)^{2x} - (a^2)^x}{\sqrt{a^x a^{3x}}}$ where $a > 0$

e) $\frac{2^{3x} + \sqrt{9 \cdot 6^{4x}}}{4^x}$

10. A scientist is growing some bacteria in a petri dish. Suppose the petri dish starts with 100 bacteria, and the number of bacteria doubles each hour. Let $B(t)$ be the number of bacteria in the petri dish after t hours.
- a) Write a formula for $B(t)$.
- b) What is the percent change per hour in the number of bacteria?
- c) Write a formula for the number of bacteria in the petri dish after d days. (Hint: if we know the bacteria multiplies by a given factor each hour, consider how many times we'd like to multiply by that factor each day.)

11. Sketch graphs of the following functions, and label all intercepts and asymptotes.

a) $f(x) = -1.5(0.6)^x$

b) $g(t) = 10 + 1.5^t$

c) $j(x) = 2 - 4^{-x}$

5.2 APPLICATIONS OF EXPONENTIALS

You should be able to:

- Model constant percent change over different intervals (like “decreases by 30% every 12 hours” or “triples every 4 days”), including compound interest.
- Explain what *half-life* means, and model phenomena that have a half-life.
- Reason about how an exponential function grows or decays over different time intervals. (For example, if a quantity grows by 10% every year, will it grow by exactly 20%, more than 20%, or less than 20% in 2 years?)

1. Last time, we talked about exponential functions; what was special about them?

2. **Warm up.**

a) If the population of a country after t years is $5(1.07)^t$ million people, by what percent is the population growing each year?

b) Generalize: if the population after t years is instead $f(t) = C \cdot b^t$, what's the percent change per year?

3. (Problem Set 22, #3)

Each week, Lallemand starts with 0.00005 pounds of yeast; the amount of yeast doubles every 4 hours.

a) Write a formula that gives the number of pounds of yeast t hours after the start of the week.

b) Write a formula that gives the number of pounds of yeast s days after the start of the week.

4. **Finance (compound interest).**

a) Marius decides to open a bank account with an opening deposit of \$1000. Suppose that the account earns a nominal annual interest rate of 6%, compounded annually.⁽¹⁾ Assuming Marius completely ignores the account after he opens it (so he doesn't make any deposits or withdrawals), how much money does the account have t years after Marius opens it?

b) After 2 years, by how much has Marius's balance increased?

- i. 12%
- ii. More than 12%
- iii. Less than 12%

Why?

⁽¹⁾The word "nominal" means "in name only". For economists, there are two uses of the phrase "nominal interest rate"; one is trying to account for the effects of inflation. We're using it in the other sense, which will be easier to understand after you do (c).

- c) Suppose that Marius had instead deposited his money in a bank that offered a 6% nominal annual interest rate, compounded *quarterly* (4 times a year). This means that, each quarter, the account earns _____% interest. How much money would the account have after t years?

In this account, what is the percent increase in money per year? (This is called the **annual percentage yield** of the account.)

- d) What if Marius had used a bank that offered 365 compounding periods a year? How much money would the account have after t years, and what would the annual percentage yield be?

5. **Chemistry (radioactive decay).** The half-life of the radioactive isotope radium-226 is approximately 1600 years. This means that, every 1600 years, the amount of radium-226 _____.

- a) Suppose you have a sample of radium-226 whose size in mg is currently S_0 . Write a formula for the amount of radium-226 that remains in the sample t centuries from now.

✓ What's a quick way to check that your answer is reasonable?

- b) Sketch a graph of the amount of radium-226 in the sample as a function of time.

- c) Every exponential function can be written in the form $C \cdot b^t$; what are C and b for the exponential function you wrote down in (a)?

- d) Suppose you have a sample of radium-226. By what percent does the amount of radium-226 in the sample change per century?

e) How many years would it take a 100 mg sample of radium-226 to decay to 25 mg?

f) How many years would it take a 100 mg sample of radium-226 to decay to 10 mg?

6. **Sociology.** The population in a certain area of the country is increasing. In 1990 the population was 100,000; by 2010, it was 200,000.

a) If the population increased linearly between 1990 and 2010, was the population in 2000 equal to 150,000, greater than 150,000, or less than 150,000? Explain your reasoning with a picture.

b) Suppose instead that the population increased exponentially between 1990 and 2020.

i. Under this assumption, was the population in 2000 equal to 150,000, greater than 150,000, or less than 150,000? Explain your reasoning with a picture.

ii. Under this assumption, was the population in 2020 equal to 250,000, greater than 250,000, or less than 250,000? Explain.

7. In each part, you're given a function. Do the following two things:

- Sketch its graph. Label all intercepts and asymptotes.
- Decide whether the function is exponential. If so, it can be written as $f(t) = C \cdot b^t$; identify C and b .

a) $f(t) = 4 \cdot 2^{-t/4}$

b) $f(t) = \left(\frac{1}{3}\right)^{t+1} - 3$

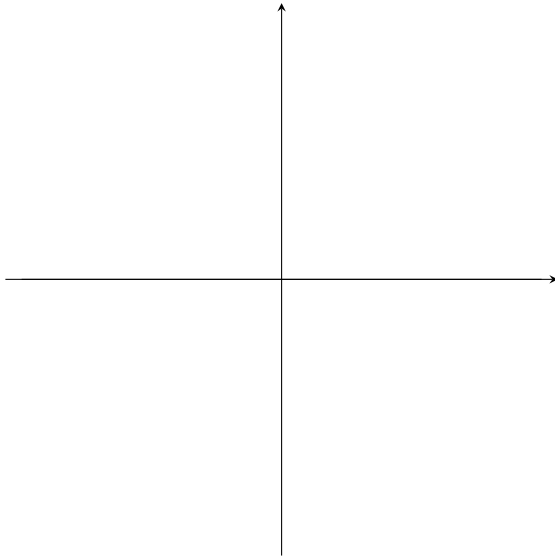
c) $f(t) = 5^{t-2} \cdot 6^{-t/2}$

5.3 DERIVATIVES OF EXPONENTIAL FUNCTIONS

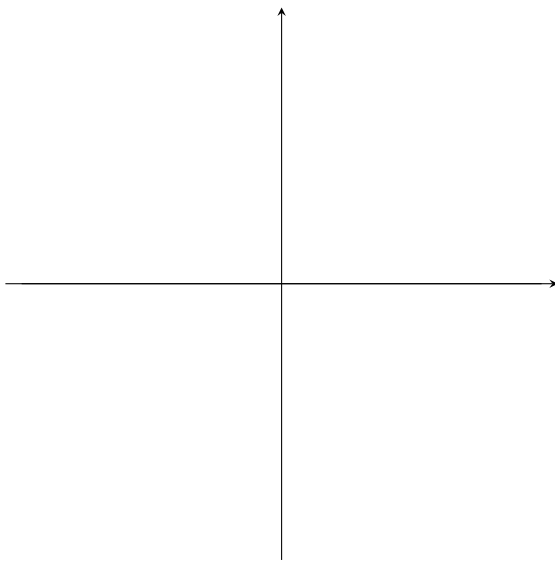
You should be able to:

- Explain what we mean when we say “the derivative of an exponential function is proportional to the function itself”.
- Explain how the number e is defined in terms of derivatives.
- Differentiate e^{kx} where k is any constant.

1. a) **Warm-up.** Sketch the graphs of $f(x) = 2^x$ and $g(x) = 3^x$ on the axes below, and label all intercepts.



- b) Use your graphs of f and g to sketch graphs of f' and g' on the axes below. While your graph won't be precise, you should be able to determine key characteristics (positive/negative, increasing/decreasing, etc.).



2. a) In workshop, you came up with something like the following estimates.

$f(x) = 2^x$			$g(x) = 3^x$			$h(x) = 10^x$		
$f'(0)$	$\approx$	0.693	$g'(0)$	$\approx$	1.10	$h'(0)$	$\approx$	2.30
$f'(1)$	$\approx$	1.39	$g'(1)$	$\approx$	3.30	$h'(1)$	$\approx$	23.03
$f'(2)$	$\approx$	2.77	$g'(2)$	$\approx$	9.89	$h'(2)$	$\approx$	230.3
$f'(3)$	$\approx$	5.55	$g'(3)$	$\approx$	29.66	$h'(3)$	$\approx$	2302.6
$f'(4)$	$\approx$	11.09	$g'(4)$	$\approx$	88.99	$h'(4)$	$\approx$	23026

What patterns did you notice? Can you guess formulas for $h'(x)$, $g'(x)$, and $f'(x)$?

- b) ✓ *Consistency check!* Are the formulas you guessed for $f'(x)$ and $g'(x)$ consistent with the graphs you sketched in #1?

3. Now, let's show that our guess for the derivative of $f(x) = 2^x$ is correct!

- a) Using the limit definition of the derivative, write a limit that's equal to $f'(0)$.

- b) Using the limit definition of the derivative, write a limit that's equal to $f'(x)$. Does this confirm your guess? (Is there some algebra you can do to rewrite the limit?)

The Derivative of Exponential Functions (first observation!)

If $f(x) = b^x$ (where b is a positive constant), then $f'(x) = \underline{\hspace{2cm}}$.

4. What can you say about $\frac{d}{dx}(2.5^x)$?

The number e . We define e to be the number b such that the slope of $y = b^x$ at $x = 0$ is ____.

5. a) What does the definition of e tell you about the derivative of e^x ?

- b) What can you say about the size of e ?

6. Now, let's find the derivative of $g(x) = e^{3x}$!

- a) $g(x) = e^{3x}$ can be written as b^x for what value of b ? What does that tell you about $g'(x)$?

- b) Let $f(x) = e^x$. How does the graph of $g(x)$ relate to the graph of $f(x)$? What does that tell you about how $g'(0)$ relates to $f'(0)$?

- c) What's $g'(x)$? (We can also confirm this using the Chain Rule!)

The Derivative of e^x and e^{kx}
 $\frac{d}{dx}e^x = \underline{\hspace{2cm}}$. More generally, for any constant k , $\frac{d}{dx}e^{kx} = \underline{\hspace{2cm}}$.

7. General exponential functions.

- a) We've now seen that $\frac{d}{dx}(2^x) = (\approx 0.693)2^x$, $\frac{d}{dx}(3^x) = (\approx 1.099)3^x$, and $\frac{d}{dx}(10^x) = (\approx 2.305)10^x$.
 If $j(x) = 7 \cdot 3^x$, what's $j'(x)$? How does it relate to $j(x)$?

- b) **Generalize:** if $j(x)$ is any exponential function $j(x) = C \cdot b^x$, how does $j'(x)$ relate to $j(x)$?

We say that two functions $A(x)$ and $B(x)$ are proportional to each other if one is a constant multiple of the other; in other words,

$$A(x) = kB(x)$$

for some constant k (called the *constant of proportionality*). So, **the derivative of an exponential function is proportional to the function itself!** Moreover, exponential functions are the only functions with this property!

8. Find the derivative of each function.

a) $f(x) = \frac{x^2 e^{3x}}{7}$

b) $f(x) = \frac{e^{5x} + 1}{e^{2x}}$

c) $f(x) = \frac{e^{2x}}{e^{5x} + 1}$

9. Let $f(x) = 2ex - e^{2x}$.

a) Where is f increasing? Where is f decreasing? How do we determine this?

- b) Where is f concave up? Where is f concave down? How do we determine this?
- c) Does f achieve a global maximum on $(-\infty, \infty)$? If so, what is the maximum value? How about a global minimum?
- d) Sketch a rough graph of f .
- e) What are $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$?

5.4 MODELING WITH EXPONENTIAL FUNCTIONS

You should be able to:

- Fit an exponential function to two data points, a graph, or a verbal description.
- Apply the fact that the derivative of an exponential function is proportional to the function.

1. *Warm up.* Evaluate the following.

a) $\lim_{x \rightarrow \infty} 4 \left(\frac{1}{3} \right)^x$

b) $\lim_{x \rightarrow -\infty} 4 \left(\frac{1}{3} \right)^x$

c) $\lim_{x \rightarrow \infty} 5 - \left(\frac{4}{3} \right)^{-x}$

d) $\lim_{x \rightarrow -\infty} 5 - \left(\frac{4}{3} \right)^{-x}$

2. A rabbit population is growing exponentially. On June 1, there are 900 rabbits. 14 days later, there are 1200 rabbits. We'd like to write a formula for $R(t)$, the number of rabbits t days after June 1. Here are two approaches.

- a) **Approach 1.** From the given data, we can see that the rabbit population gets multiplied by ____ every ____ days, so $R(t) = \underline{\hspace{2cm}}$.

From previous page: A rabbit population is growing exponentially. On June 1, there are 900 rabbits. 14 days later, there are 1200 rabbits. We'd like to write a formula for $R(t)$, the number of rabbits t days after June 1.

- b) **Approach 2.** Since the population is growing exponentially, we can write $R(t) = C \cdot b^t$; we need to find C and b .

What are $R(0)$ and $R(14)$? How can you use these two facts to find C and b ?

- c) Which approach do you prefer?

3. Find an exponential function through the points $(4, 10)$ and $(6, 15)$. (Whenever possible, it's helpful to identify what one output is multiplied by to get the other output.)

✓ *How can you check your answer?*

- The derivative of a linear function $f(x)$ is ...
- The derivative of an exponential function $f(x)$ is...

4. A population of bacteria in a petri dish is growing exponentially. When there are 8000 bacteria, the population is growing at a rate of 400 bacteria per hour.

- a) How quickly (in bacteria/hr) is the bacteria population growing when there are 10,000 bacteria in the dish?

b) If there were 1000 bacteria at time $t = 0$ (with t in hours), which of the following is a correct formula for $P(t)$, the number of bacteria at time t ?

i. $P(t) = 1000e^{20t}$

ii. $P(t) = 1000e^{t/20}$

iii. $P(t) = 50e^t$

5. The function $f(x) = \frac{3x}{e^x}$ is an example of a surge function. Surge functions are used to model many situations. For instance, the amount of an IV drug in the blood x minutes after the drug is administered can be modeled as a surge function.

Our ultimate goal in this problem is to sketch the graph of f .

- a) We can evaluate one of $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$. Which one? Do you have a guess as to what the value of the other one is?

- b) Where is f increasing? Where is f decreasing?

- c) Where is f concave up? Where is f concave down?

- d) Find all intercepts of f .
- e) Use your answers to the previous parts to sketch a graph of $f(x)$. Label asymptotes, intercepts, and the x -values of turning points and inflection points.
- f) Does f have a global maximum on $(-3, 3)$? If so, where? How about a global minimum?

6. Suppose a, b are numbers such that $3^a = 10$ and $3^b = \frac{1}{7}$.

a) In each statement, fill in the blanks with consecutive integers (for example, 3 and 4).

- a is between ____ and ____.
- b is between ____ and ____.

b) Simplify the following expressions. (Your answers shouldn't involve a , b , or c .)

i. 3^{a+b}

ii. 3^{-b}

iii. $3^{2a-b/4}$

iv. 3^{-4a}

5.5 EXAM 2 REVIEW

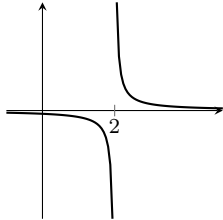
Material for in-class activities for Wednesday, November 12.

1. *You be the grader!*

a) A Math Ma student is working on the following problem.

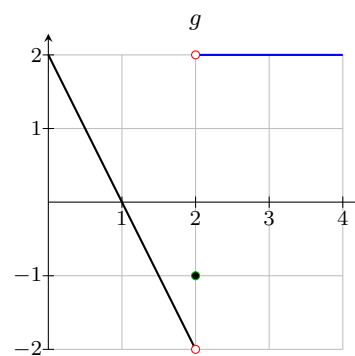
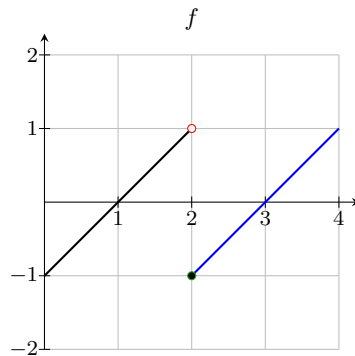
Sketch an example of a function $f(x)$ such that $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$ both exist but $\lim_{x \rightarrow 2} f(x)$ does not exist.

The student submits the answer below. What feedback would you give them?



b) Another student is working on this problem.

Given the graphs of f and g below, find $\lim_{x \rightarrow 2} f(x)g(x)$ and $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$.



The student writes:

Since $\lim_{x \rightarrow 2} f(x)$ DNE, $\lim_{x \rightarrow 2} f(x)g(x)$ DNE. Since $\lim_{x \rightarrow 1} f(x) = 0$, $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 0$.

What feedback would you give the student?

2. Q-Tips[®] are a brand of 3-inch long cotton swabs. The manufacturer would like to sell them in boxes with volume 37.5 cubic inches. One side of the box is made from cardboard, and the other five sides are made of plastic; to fit the swabs, one edge of the cardboard must be 3 inches long. What dimensions should the box be to minimize the amount of plastic used?

3. *E. coli* is a bacteria sometimes found in ground beef that can cause fatal food poisoning. Naturally, food safety inspectors would like to be able to test for it; the challenge is to be able to take a tiny sample of ground beef (so that the rest can be sold to customers) and check for *E. coli*, even if the sample has just a single bacterium.

DNA tests aren't sensitive enough to detect just a single bacterium, so food inspectors put the ground beef sample in "a broth infused with nutrients that *E. coli* likes to eat... to rest for 10 hours". Let's understand why. Under such conditions, it's estimated that a population of *E. coli* doubles every 30 minutes.

- a) How many *E. coli* are there t minutes after a single bacterium is put in the broth?
- b) How many *E. coli* are there h hours after a single bacterium is put in the broth?
- c) ✓ What kind of check can you do to make sure your answers are reasonable?
- d) How many bacteria are there after 10 hours? Use the fact that $2^{10} \approx 1000$ to estimate roughly how big this number is.
- e) By what percent does the number of bacteria in the broth grow per hour?

4. Let $g(x) = x^3 - 12x$.

a) Is g even, odd, or neither? Justify your answer.

b) Find all critical points of g , and classify each as a local minimum, local maximum, or neither.

c) What's the y -intercept of g ? Is it easy to find the x -intercepts? If so, find them.

d) Sketch a rough graph of $g(x)$. You need not depict the concavity accurately.

Let $f(x) = |g(2x)|$.

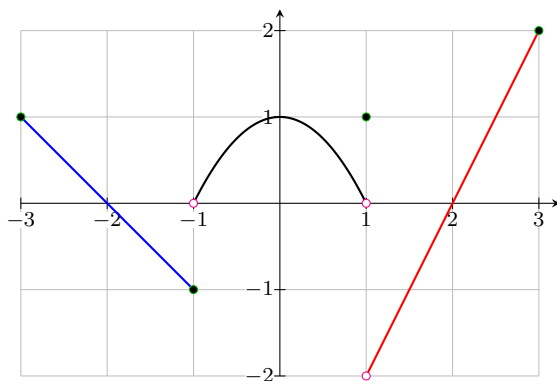
e) Sketch a rough graph of $f(x)$. Label any intercepts.

f) Where (at what x -values) are the local minima of f ? Where are the local maxima?

g) Find the global maximum and minimum of f on $[-2, 2]$. Please tell us both the global maximum / minimum value (the y -value) and where (at what x -value(s)) each occurs.

5. Let $g(x) = 2^{x-3} \cdot 5^{-x/2}$. Rewrite $g(x)$ in the form $g(x) = C \cdot b^x$, and sketch its graph. Label all intercepts and asymptotes.

6. Here's the graph of the function f and a formula for a function g .



$$g(x) = \begin{cases} 2^x - 1 & \text{if } x < -2, \\ x^2 - 1 & \text{if } -2 \leq x < 1, \\ 2 & \text{if } x = 1, \\ 4 - x^2 & \text{if } x > 1. \end{cases}$$

Evaluate each of the following.

a) $\lim_{x \rightarrow 1} f(x)g(x)$

b) $\lim_{x \rightarrow -1} f(x)g(x)$

c) $\lim_{x \rightarrow 1^-} \frac{g(x)}{x^2 - 4x + 3}$

d) $\lim_{x \rightarrow 1^+} \frac{g(x)}{x^2 - 4x + 3}$

e) $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3}$

f) $\lim_{x \rightarrow -2} \frac{g(x)}{f(x)}$

UNIT 6

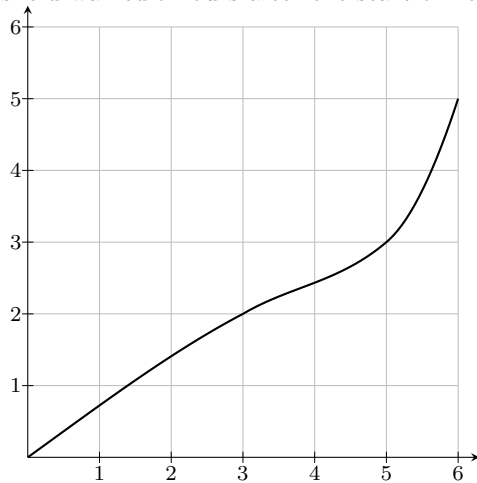
INVERSE FUNCTIONS AND LOGARITHMS

6.1 INVERSE FUNCTIONS

You should be able to:

- Determine whether a function f is invertible based on a graph, description, or formula. If so, determine the domain and range of f^{-1} .
- If f is an invertible function given verbally (for example, if a farmer is growing a giant pumpkin and $f(t)$ gives the pumpkin's weight at time t), describe what f^{-1} represents.
- Find a formula for f^{-1} when given a formula for an invertible function f .
- Sketch the graph of f^{-1} from a graph of an invertible function f .

1. Last weekend, Shay went for a long leisurely walk around Boston. Here's the graph of $f(t)$, the distance she'd walked t hours after the start of her walk.



a) How long did it take Shay to walk 2 km?

b) How long did it take Shay to walk 3 km?

It's natural to want a function that takes as input a distance and gives as output the amount of time it took Shay to walk that distance. We call this the inverse function of f and write it as f^{-1} and read this as " f inverse." While an inverse function is a noun, the adjective form is invertible. Using these in context, we'd say " f is invertible and its inverse function is f^{-1} ."

c) What do your answers to (a) and (b) tell you about f^{-1} ?

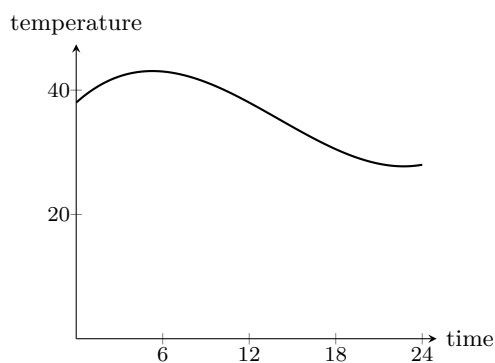
In general, if g is a function, then g^{-1} is the function that _____.

d) What are the domain and range of f^{-1} ? How do they relate to the domain and range of f ?

e) Can you sketch the graph of f^{-1} on the above axes?

- f) In general, if g is a function, how does the graph of g^{-1} relate to the graph of g ?
- g) Remember that, if a and b are two functions, we call $a(b(x))$ the composition of a and b . What happens if you compose f^{-1} and f ? How about in the opposite order?

2. Suppose the following graph shows the temperature at Logan Airport over a particular 24-hour period. Let $f(t)$ be the temperature at time t . What's $f^{-1}(40)$?



3. In this problem, we'll look at the two functions $a(x) = x^2$ and $b(x) = x^3$.
- a) Is each function invertible? If so, find a formula for its inverse. If not, explain why not.

You should have found that one of the two functions is invertible. Let's call the invertible function f and the non-invertible function g .

b) Graph $f(x)$ and $f^{-1}(x)$ on the same set of axes.

c) What's $(f(x))^{-1}$? Is it the same as $f^{-1}(x)$?

d) Can you restrict the domain of $g(x)$ to make it invertible? Sketch the inverse of your restricted g . What's a formula for the inverse of your restricted g ?

4. **Determining invertibility from a formula.** Decide whether each of the following functions is invertible. *Hint:* You know how to graph one of these functions. For the one you don't know the graph of, can you figure out where it's increasing / decreasing? How is that helpful?

a) $f(x) = 2^x - 1$

b) $h(x) = \frac{1}{2e^{3x}} - x^3$

5. One of the following two functions is invertible. Which one? Find the domain and range of the inverse function, as well as a formula for the inverse function.

a) $f(x) = -2|x - 3|$

b) $g(x) = -\frac{5}{x-2} + 3$

6. **Reasoning about a function and its inverse.** The Wonderful Widget Company manufactures wonderful widgets. The company's costs in a given month depend on how many widgets they manufacture that month. Let $g(x)$ be their monthly costs, in dollars, if they produce x widgets.

a) Is g likely to be invertible? Why or why not? If it is, what is its input and what is its output?

b) What does $g(400)$ represent? What are its units?

c) If g is invertible, what does $g^{-1}(400)$ represent? What are its units?

d) What does $g'(400)$ represent? What are its units?

It turns out that the Wonderful Widget Company has fixed monthly operating costs of \$10,000, and it costs them an additional \$7 to manufacture each widget.

e) Using this information, write a formula for $g(x)$.

f) Now, write a formula for g^{-1} .

Summarizing what we know about inverse functions!

- Conceptually, describe an inverse function in your own words (with as little technical language as possible):
- If we're given a graph of a function, how do we determine whether the function is invertible? How do we draw the graph of the inverse function?
- If we're given the algebraic representation of a function (a formula), how do we determine the formula for its inverse (assuming one exists)?

6.2 AN INTRODUCTION TO LOGARITHMS

You should be able to:

- State and apply the definition of a logarithm, and explain how logarithms are related to exponentials.
- Explain how to simplify $\log_b(b^{\star})$ and $b^{\log_b \heartsuit}$.
- Sketch $y = \log_b x$ and explain how the graph and its important features (domain, range, asymptotes, intercepts, etc.) relate to the graph of $y = b^x$.
- Differentiate b^x for any constant $b > 0$.

1. When a new technology (like iPhones, Facebook, Twitter, etc.) starts to become popular, it often experiences exponential growth. In 2007, Facebook reported that the number of Facebook members was growing by about 3% per week. On October 25 of that year, Facebook reached 50 million members; if the growth trend continued, when did the number of members reach 100 million?

2. Approximating the numerical value of logarithms

- a) We read $\log_7(40)$ as “_____.”

This number is the power to which _____ must be raised in order to get _____.

Find upper and lower bounds for this number: _____ $< \log_7(40) <$ _____.

- b) Evaluate the following.

$$\log_7(49) = \underline{\hspace{2cm}} \qquad \log_7\left(\frac{1}{\sqrt{7}}\right) = \underline{\hspace{2cm}} \qquad \log_7(1) = \underline{\hspace{2cm}}$$

Make up two more examples, one where the log is positive and one where it's negative:

$$\log_7(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}} \qquad \log_7(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$$

c) If $\log_7(v) = w$, then _____ is the power to which _____ must be raised in order to get _____. This can be rewritten as an exponential equation _____.

d) If $f(x) = \log_7(x)$, find $f^{-1}(x)$.

e) Use your answer to (d) to graph $y = \log_7(x)$, and label at least 3 points on your graph.

What's the domain of $\log_7(x)$?

Where are its intercepts?

f) What's $\log_7(7^\star)$? _____

What's the range of $\log_7(x)$?

Does it have any asymptotes?

What's $7^{\log_7(\heartsuit)}$? _____

Notation for a special base. We call the base e logarithm the “natural logarithm” and write $\ln(x)$ as shorthand for $\log_e(x)$. So, $\ln(x)$ is the power to which _____ must be raised in order to get _____.

3. Make up a few examples, including one where the natural log is positive and one where it's negative:

$\ln(\text{_____}) = \text{_____}$ $\ln(\text{_____}) = \text{_____}$ $\ln(\text{_____}) = \text{_____}$

4. Usually, when we have a function f , it is not considered acceptable to write fx instead of $f(x)$. However, with logarithmic functions, it is! So, $\log_7 40$ is a common notation for $\log_7(40)$, and $\ln(5)$ is commonly written as $\ln 5$.

However, it **isn't** okay to write an expression like $\log_5 x^2$ because it's ambiguous. What are 2 ways a reader might interpret $\log_5 x^2$? What would each interpretation equal if $x = 5$?

⚠ In different sources, you may encounter the notation $\log(x)$, where no base is written. This doesn't have a single set meaning! In the Gottlieb textbook, this means $\log_{10}(x)$. But in other sources, it might mean $\log_2(x)$ or $\log_e(x)$. We'll avoid this notation in Math Ma.

5. Simplify the following expressions. (One is already simplified!)

a) $7^{\log_7 49}$

b) $e^{\ln 2}$

c) $5^{\log_5 6 + \log_5 7}$

d) $6^{3 \log_6 10}$

e) $3^{\log_7 3}$

f) $e^{-\ln 3}$

6. In #1, the number of millions of Facebook users t weeks after October 25 was $50(1.03^t)$.

a) Can you write 1.03 as e^k for some k ? If so, what's k ?

b) Rewrite $50(1.03^t)$ in the form Ce^{kt} .

Every exponential function can be written as $f(x) = \underline{\hspace{2cm}}$ or as $f(x) = \underline{\hspace{2cm}}$.

7. Since every exponential function f can be written as $f(x) = C \cdot b^x$ and as $f(x) = Ce^{kx}$, rewrite each function below in both forms.

a) $f(x) = 5^{2x-1}$

b) $g(x) = 7e^{3x+2}$

8. a) Sketch the graph of $f(x) = 4 - \log_{10} x$. What are the intercepts and asymptotes of f ?

b) Label at least three points on the graph.

c) Is f invertible? If so, find the domain and range of f^{-1} , as well as a formula for $f^{-1}(x)$.

6.3 LOGARITHM RULES

You should be able to:

- Apply basic properties of logarithms. (Can we simplify the log of a product? How about the log of a sum, quotient, or power?)
- Change bases of logarithms (for example, rewrite $\log_2 3$ in terms of $\ln$ or $\log_7$).

Warm up! Translate the statement $\log_2(3) = x$ into words.

1. Simplify each as much as possible, if you can. If it is already simplified or undefined, say so.

a) $\log_5 25 = \underline{\hspace{1cm}}$ because

b) $\ln(e\sqrt{e}) = \underline{\hspace{1cm}}$ because

c) $\ln\left(\frac{1}{e\sqrt{e}}\right) = \underline{\hspace{1cm}}$ because

d) $11^{\log_5 11}$

e) $7^{\log_7 3}$

f) $e^{-3 \ln 2}$

g) $\log_3(3^5 \cdot 3^8)$

h) $\log_{10}(10^2 - 10^3)$

i) $\log_2\left(\frac{2^{11}}{2^6}\right)$

2. Here are two facts: $\log_3 4 \approx 1.26$ and $\log_3 7 \approx 1.77$. We'll use these two facts to help you approximate the following quantities, and explain your reasoning in each.

a) Rewrite $\log_3 4 \approx 1.26$ and $\log_3 7 \approx 1.77$ as relationships between quantities that involve exponential functions.

b) By substituting these expressions for 7 and 4, simplify the following:

i. $\log_3 \left(\frac{7}{4} \right)$

ii. $\log_3 28$

iii. $\log_3 (7^{2.1})$

Generalize the rules from the previous problem:

Logarithm Rules.

1.

2.

3.

3. Is there a nice rule for taking $\log_b$ of a sum? Come up with a convincing example to support your answer.

4. a) Express $\ln(wu^2)$ in terms of $A = \ln u$ and $B = \ln w$. (Begin by rewriting the relationship between the variables using exponential functions. Which base should you use?)

b) Express $\ln [(wu)^2]$ in terms of $A = \ln u$ and $B = \ln w$. Is this the same as or different from (a)?

5. If possible, simplify each expression, and write it in terms of $A = \log_2 u$ and $B = \log_2 w$ (with no logarithms in your final answer).

a) $7 \log_2 \left(\frac{8}{\sqrt{uw}} \right)$

b) $7 \log_2 (u + \sqrt{8w})$

c) $7 \log_2 \left(\frac{8}{\sqrt{uw}} \right)$. Is this the same as or different from (a)?

6. Simplify the following, if possible:

a) $\ln(7^3) - \ln(7)$

b) $(\log_3 7)(\log_3 8)$

c) $\log_8(x+1) + \log_8\left(\frac{1}{x+1}\right)$

7. On the same set of axes, graph $\ln x$, $\log_2 x$, and $\log_3 x$ in 3 different colors. (What's our usual strategy for graphing a log?) Where are the asymptotes and intercepts of each graph?

8. Rewrite $\log_5 17$ in terms of natural logs. (Strategy: Give $\log_5 17$ a name like x , rewrite the equation $x = \log_5 17$ as an equation involving exponents, and then take the natural log of both sides and solve for x .)

Change of base to natural log: $\log_b a =$

9. Explain how the change of base formula is consistent with your graphs in #7.

10. Graph $f(x) = \log_3 \left(\frac{9}{x^2} \right) + \log_3 x$. (*Hint:* First use log rules to simplify.) Label all intercepts and asymptotes, and label at least two other points on the graph with exact coordinates.
11. a) Simplify $\log_8 32$. (*Hint:* If you're having trouble, try using the strategy of #8 to rewrite this in terms of base 2 logarithms.)
- b) ✓ How can you check your answer?
12. Can you use exponent rules to simplify $e^{4 \ln 3}$? Can you use log rules to simplify $e^{4 \ln 3}$?

6.4 THE DERIVATIVES OF LOGARITHMIC AND EXPONENTIAL FUNCTIONS

You should be able to:

- Differentiate logarithmic functions, and apply this together with the differentiation rules we've learned before.
- Use algebra to rewrite functions involving exponentials or logarithms so that they can be differentiated.

Every exponential function can be written as $f(x) = \underline{\hspace{2cm}}$ or as $f(x) = \underline{\hspace{2cm}}$.

1. **Warm up.** We can use the boxed fact to find the derivative of 7^x !

a) So far, what do we know about $\frac{d}{dx} 7^x$?

b) What's $\frac{d}{dx} e^{kx}$?

c) Rewrite 7^x in the form Ce^{kx} , and use this to find $\frac{d}{dx} 7^x$.

The Derivative of b^x : If $b > 0$, then $\frac{d}{dx} b^x =$ _____

2. Let's turn our attention to logarithms.

a) In white chalk, sketch the graph of $f(x) = \ln x$. On the same set of axes, sketch the graph of $g(x) = \log_5 x$ in colored chalk.

b) In white chalk, sketch the graph of $f'(x)$. On the same set of axes, sketch the graph of $g'(x)$ in colored chalk.

c) Which is greater, $\ln 10$ or $\log_5 10$? Why?

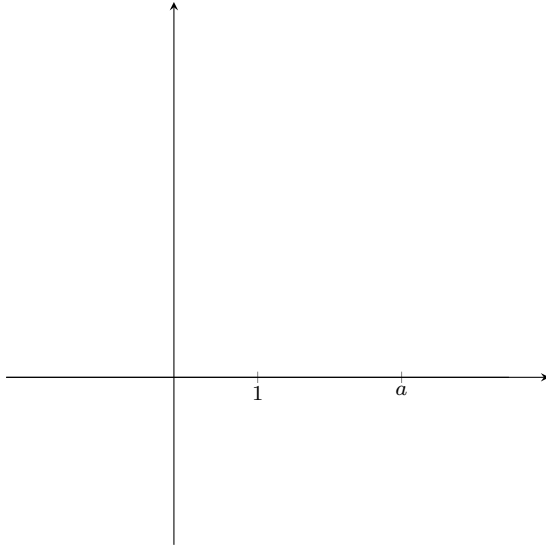
d) Use the change of base formula to express $g(x) = \log_5 x$ in terms of $f(x) = \ln x$. Is this consistent with your graphs?

What does this tell you about how $g'(x)$ relates to $f'(x)$?

3. Let $f(x) = \ln x$. In this problem, we'll figure out a formula for $f'(a)$.

a) Write the limit definition of $f'(a)$.

b) On the axes below, a value a is marked. Sketch the graph of $f(x) = \ln x$ together with a line whose slope is $f'(a)$.



c) On the picture above, sketch the reflection of both the curve and the line over $y = x$.

d) Find the slope of the reflected line. (*Hint:* What's the reflected function?)

e) What does that tell you about the slope of the original line?

f) What's $f'(x)$? Is this consistent with the graph you sketched in #2(b)?

4. Use #2(d) and #3(f) to find the derivative of $\log_5 x$.

Derivatives we know.

• $\frac{d}{dx}(x^n) =$ (n is a constant)	• $\frac{d}{dx}(b^x) =$ ($b > 0$ and constant)
• $\frac{d}{dx}(e^x) =$	• $\frac{d}{dx}(\ln x) =$
• $\frac{d}{dx}(e^{kx}) =$ (k is a constant)	• $\frac{d}{dx}(\log_b x) =$ ($b > 0$ and constant)

5. Write down the log rules we learned last time.

6. Find the derivative of each of the following functions. You'll need to either use logarithm rules or exponential rules to rewrite the function in a form that you can differentiate, or take advantage of the Chain Rule and differentiate right away. Experiment with both! (If the Chain Rule feels overwhelming today, rest assured we'll be reviewing derivatives tomorrow.)

a) $g(x) = \ln(3x^2)$

b) $f(t) = \ln(et^t)$

c) $j(x) = e^{3x-2}$ (Hint: We know how to differentiate functions of the form Ce^{kx} or $C \cdot b^x$. Can you rewrite this in one of those two ways?)

d) $y = \frac{\sqrt[3]{8 \cdot 5^x}}{7}$

e) $g(x) = \log_2 \left(\frac{2^x}{\sqrt[3]{x}} \right)$

f) $h(x) = e^{-3 \ln x}$

g) $g(w) = \frac{\log_7(w)}{7^w}$

7. **Applying this new derivative rule!** Let $f(x) = x \ln \sqrt{x}$.

a) What's the domain of f ?

b) Does $f(x) = x \ln \sqrt{x}$ have a global maximum on its domain? If so, where? How about a global minimum?

6.5 DERIVATIVE REVIEW, REVISITING THE CHAIN RULE

You should be able to:

- You should be able to apply the Chain Rule like a pro!
- You should be able to apply the Chain Rule in more general contexts, such as in a word problem or when given a composition of functions described by graphs or numerical data.
- You should understand how logarithms and the Chain Rule can be used to show that the Power Rule $\frac{d}{dx}(x^n) = nx^{n-1}$ holds for any real number n .

1. **Warm up** Fill in the first table, recalling the derivatives we've learned over the last month.

Table 1: Derivatives we knew before

$f(x) = x^n$	$f'(x) =$
$f(x) = e^{kx}$	$f'(x) =$
$f(x) = b^x$	$f'(x) =$
$f(x) = \ln x$	$f'(x) =$

Table 2: Derivatives with the Chain Rule

$h(x) = [g(x)]^n$	$h'(x) =$
$h(x) = e^{g(x)}$	$h'(x) =$
$h(x) = b^{g(x)}$	$h'(x) =$
$h(x) = \ln[g(x)]$	$h'(x) =$

2. Earlier this semester, we learned the Power Rule, which said that $\frac{d}{dx}x^n = nx^{n-1}$ for any real number n . However, we only *proved* this when n was either a positive integer or $\frac{1}{2}$. Now that we know the Chain Rule, we can finally prove the Power Rule for all real numbers n . In this problem, you'll use the Chain Rule to prove that the derivative of x^π is $\pi x^{\pi-1}$. (The same idea works for any exponent, not just π .)

a) Rewrite $f(x) = x^\pi$ as $f(x) = e^{\text{something}}$. What is the "something" in the exponent?

b) Now, use the Chain Rule and what you know about the derivative of e^x to show that $f'(x) = \pi x^{\pi-1}$.

3. Differentiate the following functions. Try to be energy-efficient, and don't operate on automatic pilot. (Don't try to simplify your answers.)

a) $h(x) = e^{3x} + x^6 \cdot \ln(7x^2 + 3)$

b) $j(u) = \ln(\ln u)$

c) $h(x) = \ln(8x^4 + x^2)$

d) $q(t) = e^{t^3} - 12e^2t$

e) $y = \frac{\sqrt{x^2 + 1}}{\pi}$

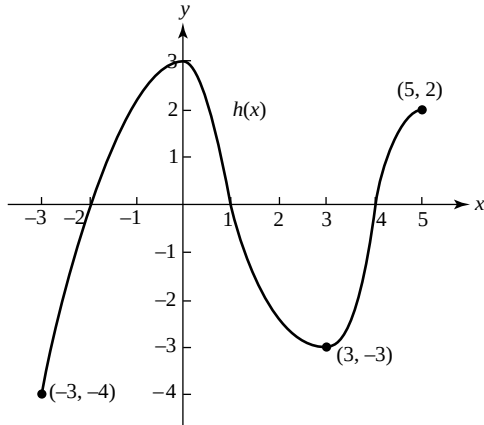
f) $f(t) = \frac{5}{\sqrt[3]{1+e^t}}.$

g) $h(x) = e^{x^2} \cdot \ln(7x^2 + 3)$

h) $j(x) = [\ln(7x^2 + 3)]^5$

4. In preparation for a party, you're blowing up spherical balloons. The volume of a sphere can be related to its radius by the formula $v = \frac{4}{3}\pi r^3$.
- a) Initially, you're just interested in the relationship between the sphere's volume and its radius. Find a formula for the instantaneous rate of change of the balloon's volume with respect to its radius. Identify its units.
- b) At what rate is the volume changing at the instant the radius expands through 5 cm? Identify the units.
- c) At this instant (when the radius is 5 cm), the radius is increasing at 2 cm per second. Given this additional information, what is the instantaneous rate of change of the balloon's volume with respect to **time** at this moment?

5. Here is the graph of a function $h(x)$.



a) For which values of x is $h'(x)$ negative?

Define a new function $f(x)$ by $f(x) = [h(x)]^2$.

d) Find all critical points of $f(x)$, and classify each as a local minimum, local maximum, or neither.
(Note: This is asking about $f(x)$, not $h(x)$!)

e) Does f achieve an absolute maximum on $[-3, 5]$? If so, where?

f) Does f achieve an absolute minimum on $[-3, 5]$? If so, where?

6.6 SOLVING EQUATIONS WITH LOGARITHMS AND EXPONENTS

You should be able to:

- Solve equations involving exponentials and logarithms.

1. A country's government is doing some demographic research. They estimate that, t years from now, the number of working-age adults will be $7(0.98)^t$ million, and the number of retirement-age adults will be $2(1.03)^t$ million. When the number of working-age adults falls to 3 times the number of retirement-age adults, they will need to put in additional measures to finance their health care system, so they would like to predict when this will happen. What equation should they solve to find this out?

In each problem, solve for x .

2. *Warm up.*

a) $6(x + 3) = 7(2x - 1)$

b) $2x^2 + 1 = -5x$

3. *3 basic types of equations.*

a) $7^x = 12$

b) $x^7 = 12$

c) $\log_7 x = 12$

d) $7^x = -12$

e) $\log_7 x = -12$

4. Now, we'll solve $7^{3x+1} = 5 \cdot 7^{x+3}$.

Method 1: Start by dividing both sides by 7^{x+3} to put the powers of 7 together:

Method 2: Since x is in the exponent, start by taking an appropriate log of both sides:

5. a) $4^{x+3} = 5^{2x-1}$

b) $2^{x^2} - 5e^x = 0$ (How is 2^{x^2} different from $(2^x)^2$? What does each one mean if $x = 10$?)

c) Solve the equation from #1.

6.  Tips for successful equation solving:

- Remind yourself what you're doing in each step. Are you taking the natural log of both sides? Raising 5 to both sides? Simplifying one side using a log rule?
- Work with equations! That is, every step you write down should have an equals sign in it.
- Remember that, if you're doing something to one side of an equation (other than just rewriting it), you have to do the same thing to the other side.
- Take it one step at a time! Most mistakes happen when you try to do multiple things at once (like taking a log *and* rewriting using log rules). It never hurts to write out more steps.
- When solving an equation that has logs in it, it's easy to introduce extraneous solutions. So, at the end, be sure to check that each solution is actually in the domain of the equation.
- If you end up with a simple answer, take an extra second to plug it in to the original equation and check whether it works.

a) $\ln(x + 5) = 2\ln(x + 3)$

Check your solutions by plugging them back into the original equation.

b) $\log_3(x+1) - \log_3 x = 2$

c) $(\log_{10} x)^2 + \log_{10} x = 6$

d) $\ln(5x^2) = 3 + \ln x$

7. a) $(\log_3 x)^3 = \log_3 x$

b) $2^{2x+3} = 8^{x-7}$ (*Hint:* There's exactly one solution, and it's an integer.)

c) $3^x \cdot \frac{5}{3^{x+1}} = 0$

$$\text{d) } 49^x - \frac{\sqrt{7}}{7^{x^2}} = 0$$

$$\text{e) } 7 \cdot 4^{x+2} - \frac{5}{3^{2x+1}} = 0$$

UNIT 7

BEHAVIOR OF FUNCTION FAMILIES

7.1 LINEAR FUNCTIONS, EXPONENTIALS, AND LOGARITHMS IN THE REAL WORLD

You should be able to:

- Assess how appropriate it is to use a linear or exponential model to model data.
- Interpret plots where an axis has a logarithmic scale.

1. **Warm up.** Sketch the graph of $f(x) = \ln x$.

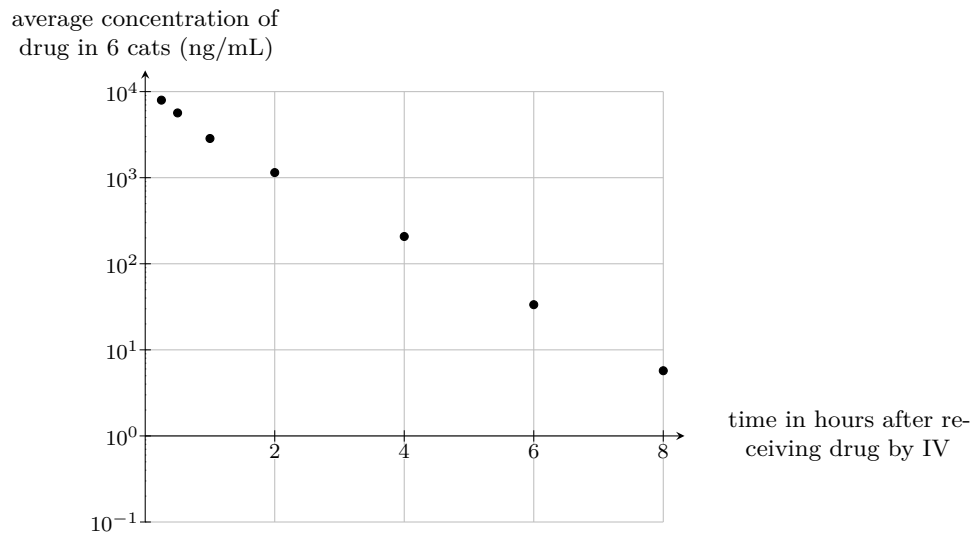
2. This article about the shoe brand Hoka says,

It took five years for Hoka's sales to accelerate from less than \$3 million to more than \$100 million. It took six more to zoom past \$1 billion.

a) Does it seem reasonable to model Hoka's sales as a linear function of time? Explain why or why not.

b) Does it seem reasonable to model Hoka's sales as an exponential function of time? Explain why or why not.

3. The following figure is reproduced from a study ⁽¹⁾ of how cats metabolize a particular chemotherapy drug, cyclophosphamide. In the study, 6 cats were given the drug by IV, and the concentration of the drug in their plasma was measured frequently afterwards.



- a) What's strange about the vertical axis?

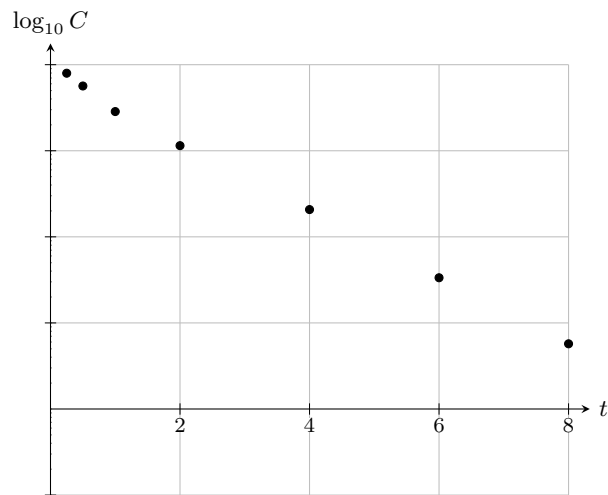
Let $C(t)$ be the average concentration of the drug in ng/mL (nanograms per milliliter) in the 6 cats t hours after being given the drug.

- b) Based on the graph, what are approximate values of $C(0)$, $C(2)$, and $C(4)$?

The data points roughly lie on a straight line; is $C(t)$ roughly a linear function?

⁽¹⁾Katherine A. Stroda et al. "Pharmacokinetics of cyclophosphamide and 4-hydroxycyclophosphamide in cats after oral, intravenous, and intraperitoneal administration of cyclophosphamide". In: *American Journal of Veterinary Research* 78.7 (2017), pp. 862-866. DOI: 10.2460/ajvr.78.7.862. URL: <https://avmajournals.avma.org/view/journals/ajvr/78/7/ajvr.78.7.862.xml>.

- c) The true interpretation of the vertical axis is that it's showing a logarithm of the concentration. In this case, since the vertical axis has powers of 10, we're really seeing a graph of $\log_{10} C$. Label the values on the vertical axis.

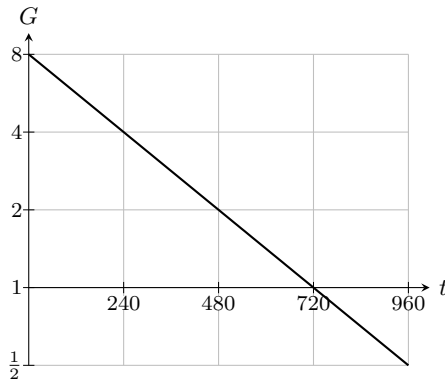


- d) Based on the graph, write an approximate formula for $\log_{10} C(t)$.

Use your formula for $\log_{10} C(t)$ to get an approximate formula for $C(t)$, and sketch a “normal” graph of $C(t)$.

- e) This type of graph is called a log-linear graph because the vertical axis is drawn using a logarithmic scale while the horizontal axis is drawn using a normal (“linear”) scale. If the log-linear graph of a function $f(x)$ is linear, what does that tell you about $f(x)$?

4. Gadolinium-153 is a radioactive isotope used in X-ray fluorescence and osteoporosis screening. Suppose you have a sample of Gadolinium-153. Because it is radioactive, the amount of Gadolinium-153 in the sample naturally decays exponentially over time. Here is a log-linear plot of $G(t)$, the number of milligrams of Gadolinium-153 in the sample after t days.

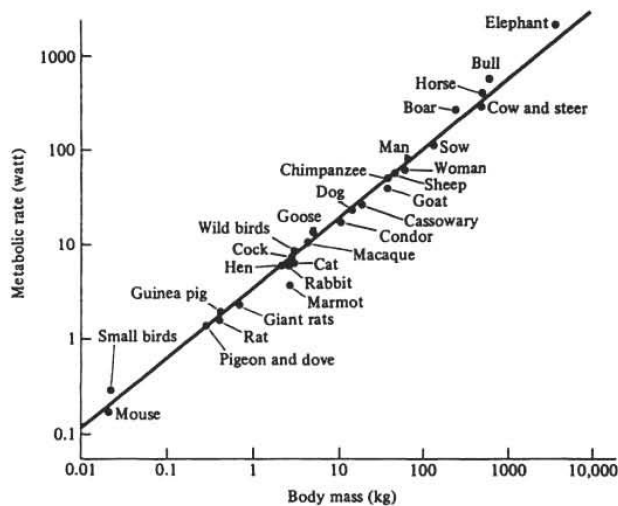


a) What's the half-life of Gadolinium-153?

b) Write a formula for $G(t)$.

c) At what time is the amount of Gadolinium-153 in the sample equal to 3 milligrams?

5. Here's a figure from a paper⁽²⁾ about how body size relates to metabolism in different animals. (This graph is known as the “mouse-to-elephant curve”.) What do you notice about the axes? Why do you think the authors of this figure have chosen to present it this way rather than using a “normal” graph?



⁽²⁾Bruce J. West. “A Fractional Probability Calculus View of Allometry”. In: *Systems* 2.2 (2014), pp. 89-118. ISSN: 2079-8954. DOI: 10.3390/systems2020089. URL: <https://www.mdpi.com/2079-8954/2/2/89>.

6. Arturo is keeping an eye on a YouTube video; at noon, he sees that it has 1200 views. 10 minutes later, it has 1400 views. Arturo decides to write two models for the number of views t minutes after noon, a linear model $L(t)$ and an exponential model $E(t)$.
- a) Sketch graphs of $L(t)$ and $E(t)$ on the same axes. For what times does the linear model predict a higher number of views?
 - b) Arturo wants to predict how quickly the number of views is increasing at 12:20 pm. Will the linear model or exponential model give a higher prediction?
 - c) Do we have enough information to decide whether a linear or exponential function is an appropriate model?
 - d) According to the linear model, how quickly is the number of views changing at 12:20 pm?
 - e) According to the exponential model, how long does it take for the number of views to reach n ? And how would you express this quantity in function notation?

f) According to the exponential model, how quickly is the number of views changing at 12:20 pm?

7. Solve the following equations.

a) $\ln(13 - 3x) - 2\ln(3 - x) = 0$

b) $2^{3x+2} - 5^{2x+7} = 0$

c) $(\ln x)^2 + 2(\ln x) + 1 = 0$

7.2 CUBICS AND HIGHER-DEGREE POLYNOMIALS

You should be able to:

- Identify the degree of a polynomial, and explain what this tells us about the possible number of roots and turning points of the polynomial.
- Describe the “end behavior” of a polynomial, and explain what this tells us about the possible number of roots and turning points of the polynomial.
- Write a formula for a polynomial if given enough information about its characteristics (for example, roots, local extrema, etc.).

A polynomial is any function that can be expressed in the form $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ where $a_0, \dots, a_n$ are constants and the exponents are all non-negative integers. The degree of a polynomial is the highest exponent in the polynomial.

1. **Polynomial or not?** Decide whether each of the following is a polynomial. If so, what is its degree?

a) $f(x) = x^2 + 18x + 3$

d) $g(t) = -et^4 + 8t - \pi$

b) $f(x) = x^{1/3} + \sqrt{x} + 5$

e) $h(w) = 5w^3 + 7w + 9w^{-1}$

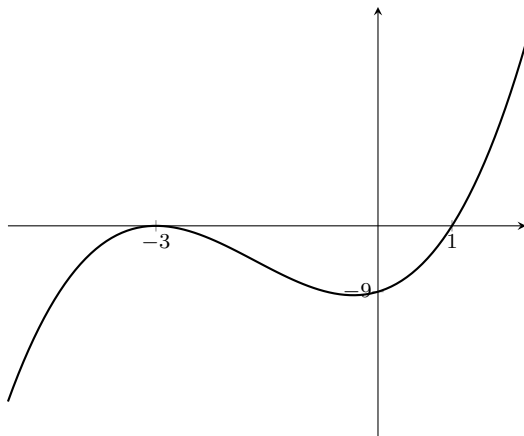
c) $f(x) = e^x$

f) $j(u) = \ln \pi$

As you know, a degree 2 polynomial is also called a *quadratic*. A degree 3 polynomial is also called a *cubic*.

2. In your last assignment, you found that the graph of $f(x) = (x - 1)(x + 3)^2 = x^3 + 5x^2 + 3x - 9$ looks like the picture below. If d is a constant, how does the graph of $g(x) = x^3 + 5x^2 + 3x + d$ relate to the graph of f ? How many roots could g have?

For each possible number of roots, sketch a possible graph of g , and say how you think the formula for $g(x)$ would factor.



3. **Comparing quadratics and cubics.** Understanding the possible ways a particular type of function can behave is important for modeling and broader reasoning. Here we'll look at characteristics across quadratics and cubics before generalizing to higher degree polynomials.

- a) If $f(x)$ is a quadratic, what can you say about $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$? (We call this the “end behavior” of f .) What if $f(x)$ is a cubic? Explain.

- b) Earlier this semester, we discussed this useful fact about functions' continuity:

Any function we can write down with a single (non-piecewise) algebraic formula is continuous at every point in its domain. In other words, the discontinuities of a function described by a single formula are where it's undefined.

Keeping this in mind, does a quadratic have any discontinuities?

- c) How many roots (x -intercepts) can a quadratic have? (Give all the possibilities.) How many roots can a cubic have? Explain.

- d) How many turning points can a quadratic have? (Again, give all the possibilities.) How many turning points can a cubic have? Explain.

- e) The graph of every quadratic can be obtained by shifting and stretching $y = x^2$. Is it true that the graph of every cubic can be obtained by shifting and stretching $y = x^3$?

4. Now we'll attempt to generalize! Suppose $f(x)$ is a degree n polynomial.

a) What can you say about the end behavior of f ? (That is, what can you say about $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?)

b) What can you say about the continuity of f ?

c) What can you say about the number of roots of f ?

d) What can you say about the number of turning points of f ?

5. Sketch the graph of $f(x) = (x - 1)(3 - x)(x - 5)$.

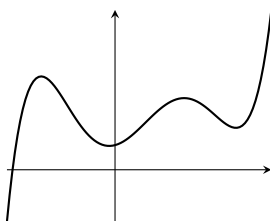
6. Find a possible formula for a degree 4 polynomial with y -intercept 4 and exactly three x -intercepts, at $x = 1, 2, 3$. How many possible answers are there?

7. In each part, give an example of a cubic function $f(x)$ with the stated properties; please give a formula. These aren't problems that you can solve just by looking at them! Instead, you'll need to come up with a strategy and then carry it out.

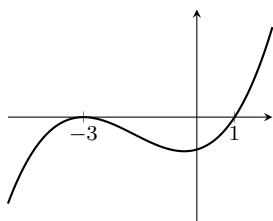
a) f has no critical points.

b) $f(x)$ has a local minimum at $x = 0$ and a local maximum at $x = 2$.

8. Here's the graph of a polynomial. What can you say about its degree? Be as specific as you can.



9. Here's the graph of $f(x) = x^3 + 5x^2 + 3x - 9$ again. The critical points of $f(x)$ are $x = -3$ and $x = -\frac{1}{3}$.



- a) Find a function $g(x)$ such that $g'(x) = f(x)$ and $g(0) = 9$.
- b) Find and classify the critical points of g .
- c) Does g achieve a global minimum on $(-\infty, \infty)$? If so, where (at what x -value), and what's the minimum value (the y -value)?
- d) Where is g concave up? Where is g concave down?
- e) Sketch the graph of g .
- f) How many roots does g have? How do you know?

7.3 COMPARING EXPONENTIALS, LOGARITHMS, AND POWER FUNCTIONS

You should be able to:

- Describe the relative growth rates of logarithmic functions, power functions (functions of the form $f(x) = x^p$ where p is a constant), and exponential functions, and use these to calculate limits.

1. Warm up.

Recall from earlier in the term: When reasoning about limits to infinity, we want to make use of two big ideas:

- that one term “eventually grows much faster than” another, which is represented by the symbol “ $\gg$ ”
- that one function “behaves like” another function, which is represented by the symbol “ $\sim$ ”.

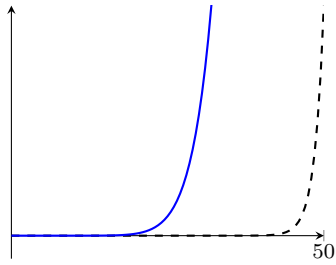
Complete reasoning, (which you’ll be asked to show in the future) requires: (1) identifying the fastest growing term in different parts of the function, (2) identifying which simplified function the given function behaves like, (3) computing the simplified function’s limit, which computes our original function’s limit.

- a) Evaluate $\lim_{x \rightarrow \infty} \frac{2x^5 + 7\sqrt{x}}{3x^6 - x^2}$, and write a careful justification with correct notation. Why is the method you used valid?

- b) What do you think $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\ln x}$ is?

2. In this problem, we'll compare the growth of $f(x) = 2^x$ and $g(x) = x^{10}$.

- a) Here are graphs of $f(x) = 2^x$ and $g(x) = x^{10}$ on $[0, 50]$. Do you think $f(x) = 2^x$ is the solid graph or the dashed graph? Why?



- b) Fill out the following table, expressing each answer as a power of 2 (like 2^7).

x	2	4	8	16	32	64	128	256
2^x								
x^{10}								

Does this change your answer to (a)?

- c) Based on your table, what do you think $\lim_{x \rightarrow \infty} \frac{2^x}{x^{10}}$ is? How about $\lim_{x \rightarrow \infty} \frac{x^{10}}{2^x}$?

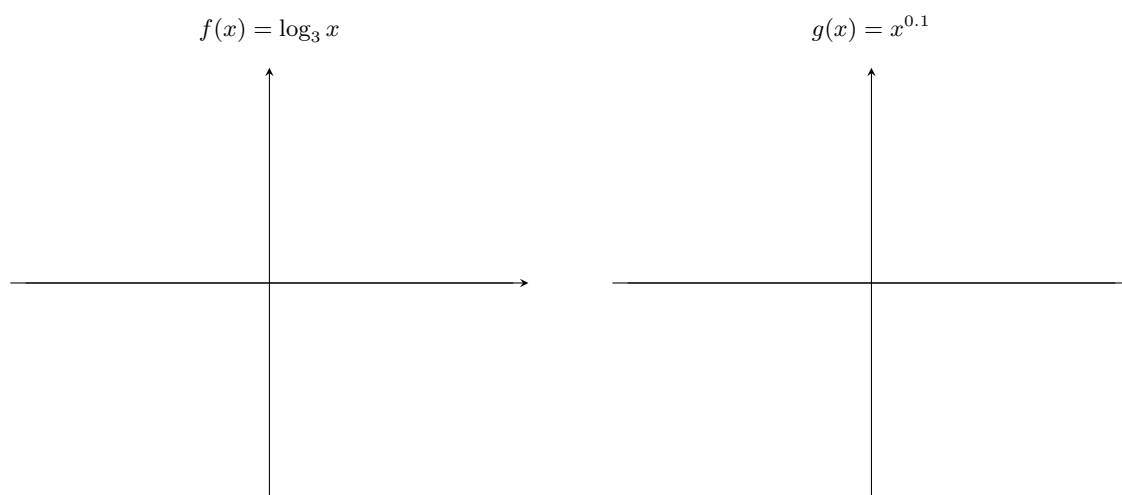
Definition. We say that $g(x)$ grows faster than $f(x)$ as $x \rightarrow \infty$ if $\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = \infty$ or, equivalently, $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$. We write this as $g(x) \gg f(x)$.

3. Tyrese would like to evaluate $\lim_{x \rightarrow \infty} \frac{1.01^x}{x^{1000}}$. He says:

I used WolframAlpha to plug $x = 1,000,000$ into $\frac{1.01^x}{x^{1000}}$; it's about 2×10^{-1679} , which is really tiny, so the limit is 0.

Assess Tyrese's reasoning.

4. a) Sketch graphs of $f(x) = \log_3 x$ and $g(x) = x^{0.1}$ on the axes below. Label at least three points on each graph with simple coordinates.



Now, we'll compare the growth rates of $f(x) = \log_3 x$ and $g(x) = x^{0.1}$.

- b) Fill out the following table.

x	3^{10}	3^{20}	3^{30}	3^{40}	3^{50}	3^{60}
$x^{0.1}$						
$\log_3 x$						

- c) Based on the table, what's $\lim_{x \rightarrow \infty} \frac{\log_3 x}{x^{0.1}}$? How about $\lim_{x \rightarrow \infty} \frac{x^{0.1}}{\log_3 x}$?

5. Fill in the blanks below with the following 3 types of functions:

- x^p where p is a positive constant
- b^x where b is a constant > 1
- $\log_b x$ where b is a constant > 1

_____ $\ll$ _____ $\ll$ _____

What about b^x with $b < 1$?

6. Which of the functions $f(x)$ below have $\lim_{x \rightarrow \infty} f(x) = \infty$? Arrange those with $\ll$ signs in between them.

$$x^{1000} \quad \log_5 x \quad \sqrt[5]{x} \quad 5^x \quad x^{-0.1} \quad 0.9^x \quad 1.1^x$$

$\triangle$ These growth rates are only relevant when thinking about what happens as $x \rightarrow \infty$!

7. With your new knowledge, do #1(b).

Remember (yes, again!): When reasoning about limits to infinity, we want to make use of two big ideas:

- that one term “eventually grows much faster than” another, which is represented by the symbol “ $>>$ ”
- that one function “behaves like” another function, which is represented by the symbol “ $\sim$ ”.

Complete reasoning, (which you’ll be asked to show in the future) requires: (1) identifying the fastest growing term in different parts of the function, (2) identifying which simplified function the given function behaves like, (3) computing the simplified function’s limit, which computes our original function’s limit.

8. Evaluate the following limits. Please write your justifications carefully.

a) $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{e^x + 3x}$

$$\text{b) } \lim_{x \rightarrow -\infty} \frac{x^3 + 2x}{e^x + 3x}$$

$$\text{c) } \lim_{x \rightarrow 0} \frac{x^3 + 2x}{e^x + 3x}$$

$$\text{d) } \lim_{x \rightarrow \infty} \frac{x^2 + 5e^{2x}}{e^{2x} - 1}$$

$$\text{e) } \lim_{x \rightarrow 0^+} \frac{x^2 + 5e^{2x}}{e^{2x} - 1}$$

$$\text{f) } \lim_{x \rightarrow \infty} \frac{\ln x + 1.01^x}{0.7^x + x^{1000}}$$

$$\text{g) } \lim_{x \rightarrow \infty} \frac{e^{2x} + x}{6e^{5x} + \log_5 x}$$

9. Let $f(x) = \frac{\ln x}{x^2}$.

- a) What's the domain of f ?
- b) Does f have any horizontal asymptotes? Justify using limits.
- c) What's $\lim_{x \rightarrow 0^+} f(x)$?
- d) Find and classify all critical points of f .
- e) Sketch a graph of f . (You need not depict the concavity accurately.)